

Hybrid AC/DC Module Technical Notebook

Current C++ Architecture, Information Models, Mathematical Models, and Solver Workflows

HACDCPF C++ Module

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Contents

1 Traffic System Analysis: A Meso-Micro Simulation Framework for Mixed Fuel and Electric Vehicles	2
1.1 Model Architecture	3
1.2 Basic Sets, Indices, and Network Definition	3
1.2.1 Time horizon	3
1.2.2 Road network	4
1.2.3 OD pairs, vehicle classes, and demand	4
1.2.4 Route and activity-plan sets	4
1.2.5 Charging stations	5
1.3 Modelling Assumptions	5
1.4 Microscopic Vehicle Model	6
1.4.1 ICV microscopic state and route choice	6
1.4.2 EV microscopic state	6
1.4.3 EV driving energy consumption	6
1.4.4 EV SOC dynamics	7
1.4.5 EV charging trigger	7
1.5 EV Economic Cost Function	7
1.5.1 EV equilibrium condition (fixed-departure special case)	9
1.6 Fleet Aggregation Model	9
1.6.1 ICV departure-rate variables and demand conservation	9
1.6.2 EV activity-plan departure-rate variables and demand conservation	9
1.6.3 Fleet-to-link inflow mapping	10
1.6.4 Fleet-to-station arrival and departure	10
1.6.5 Class-specific cumulative curves	10
1.7 CTM-Based System-Level Network Loading	10
1.7.1 Cell discretisation and CFL condition	10
1.7.2 Cell sending and receiving functions	11
1.7.3 Inter-cell flow and occupancy update	11
1.7.4 Route-commodity tracking	11
1.7.5 Inter-link node model	11
1.7.6 CTM-based Dynamic User Equilibrium (DUE)	12
1.8 LTM-Based Meso-Micro Network Loading	12
1.8.1 Cumulative curves and occupancy	12
1.8.2 Free-flow and backward-wave delays	12
1.8.3 LTM sending and receiving functions	12

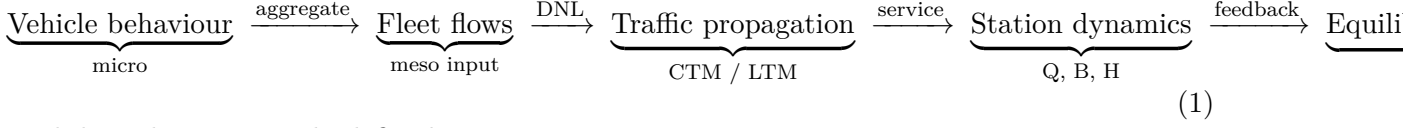
1.8.4	LTM node model	13
1.8.5	FIFO-consistent vehicle trajectory reconstruction	13
1.9	Charging Stations as Endogenous Service Nodes	14
1.9.1	Station state and dynamics	14
1.9.2	Service admission	14
1.9.3	Service completion	14
1.9.4	Station departure	14
1.9.5	Queue spillback to road	14
1.10	Charging and V2G Feedback on Traffic	14
1.11	Dynamic Equilibrium Model	15
1.11.1	Dynamic Network Loading operator	15
1.11.2	Experienced-cost operators	15
1.11.3	ICV and EV choice operators	16
1.11.4	Wardrop DUE conditions	16
1.11.5	Fixed-point equilibrium	16
1.12	Solution Algorithm: MSA for Mixed Fleet DUE	17
1.13	Key Output Metrics	17
1.14	Correctness Checklist	18
1.15	Numerical Results	18
1.15.1	Implementation diagnostics	19
1.15.2	Single-link LTM: propagation delay and CFL	19
1.15.3	Freeway sustained-capacity: LTM vs CTM throughput	19
1.15.4	CFL-violation pathology	20
1.15.5	Four-link arterial corridor	20
1.15.6	Travel-time accuracy: LTM offset vs CTM zero-delay	20
1.15.7	CTM vs LTM: model comparison	21
1.15.8	Model validation tests	21
1.15.9	EV penetration sweep	21
1.15.10	Individual EV trajectories and V2G	21
1.15.11	Fleet assignment: station-capacity benchmark (4-node, 5 EVs)	22
1.15.12	CTM-DUE MSA validation: 4-node benchmark	23
1.15.13	Joint social welfare optimizer: FormD small case (2-bus, 4 EVs)	25
1.15.14	V2G pure-LP validation (Proposition 1)	27
1.15.15	Large-scale benchmark: 10×10 grid, 10,000 EVs, IEEE-118	28
1.16	Extension Interfaces to Power and Resilience Models	30
1.17	Optimization Framework	30
1.17.1	CTM-based system-optimal traffic assignment (SO-CTM LP)	31
1.17.2	LTM-based system-optimal assignment (SO-LTM LP)	31
1.17.3	DUE as a variational inequality	32
1.17.4	Joint power-traffic social welfare optimisation	32
1.17.5	MILP for charging infrastructure design	33
1.17.6	LTM-based model predictive control (MPC)	33

1 Traffic System Analysis: A Meso-Micro Simulation Framework for Mixed Fuel and Electric Vehicles

Remark 1 (Implementation target). *This section is the mathematical contract for the `hacdcp f::evpt` traffic sub-module. Function signatures, data structures, and algorithmic steps given here are the specification that the C++ implementation in `src/ev_power_traffic/` follows. Where a function is already implemented a cross-reference to the source file is given; where it is planned, the label **Planned** appears.*

1.1 Model Architecture

The framework is organised as a five-layer pipeline:



Each layer has a precisely defined input–output contract:

- *Vehicle behaviour layer*: answers how individual vehicles choose routes and charging plans given perceived travel times, queue lengths, and energy prices.
- *Fleet aggregation layer*: converts large populations of identically-behaving vehicles into continuous OD flows, making Dynamic Network Loading (DNL) tractable.
- *CTM/LTM layer*: propagates fleet flows through the road network, computing shockwaves, spillback, travel times, and FIFO-consistent trajectories.
- *Station service layer*: models charging-station queues, plug utilisation, service completions, and dwell populations as endogenous nodes inside the network loading.
- *Equilibrium/optimisation layer*: closes the loop; at equilibrium, no vehicle has an incentive to deviate (Wardrop I for ICVs, economic UE for EVs).

This section covers:

- micro ICV and EV behaviour models (route choice, fuel consumption, SOC dynamics, charging and V2G decisions);
- fleet aggregation from micro behaviour to OD flow variables;
- CTM-based system-level network loading with cell-wise spatial state, route-commodity tracking, and DUE via MSA or variational inequality methods; system-optimal dynamic traffic assignment is formulated as an LP under triangular-FD assumptions;
- LTM-based meso-micro loading with cumulative curves, FIFO trajectory reconstruction, and an individual EV interaction layer;
- charging stations as endogenous service nodes;
- the dynamic user-equilibrium fixed-point problem and MSA solution;
- system-optimal LP, MILP infrastructure design, and MPC extensions.

This section does not cover:

- optimal power flow or grid dispatch (Section ??);
- carbon-flow or resilience analysis;
- full agent-level simulation of every vehicle.

1.2 Basic Sets, Indices, and Network Definition

1.2.1 Time horizon

The discrete time set is $\mathcal{K} = \{0, 1, \dots, T\}$ with uniform step size Δt (hours). A flow expressed in veh/h is multiplied by Δt to obtain vehicles per step.

1.2.2 Road network

The road network is a directed graph $\mathcal{G}^R = (\mathcal{N}^R, \mathcal{A}^R)$, where $\mathcal{N}^R$ is the set of road nodes and $\mathcal{A}^R$ the set of directed road links. Each link $a \in \mathcal{A}^R$ is characterised by:

$$\begin{aligned} L_a & \text{ length (km),} \\ v_a^{ff} & \text{ free-flow speed (km/h),} \\ w_a & \text{ backward wave speed (km/h),} \\ \bar{q}_a & \text{ capacity (veh/h),} \\ N_a^{jam} & \text{ jam storage (veh).} \end{aligned}$$

The triangular fundamental diagram (FD) is

$$q_a(k_a) = \begin{cases} v_a^{ff} k_a & k_a \leq k_a^c, \\ w_a(k_a^j - k_a) & k_a > k_a^c, \end{cases} \quad (2)$$

with critical density $k_a^c = \bar{q}_a/v_a^{ff}$ and jam density $k_a^j = \bar{q}_a(1/v_a^{ff} + 1/w_a)$. The backward wave speed satisfies

$$w_a = \frac{v_a^{ff} k_a^c}{k_a^j - k_a^c}. \quad (3)$$

1.2.3 OD pairs, vehicle classes, and demand

The OD set is $\mathcal{W}$; a pair $w = (o, d) \in \mathcal{W}$. The vehicle class set is $\mathcal{C} = \{\text{F}, \text{E}\}$ (fuel vehicles and EVs).

Full DUE mode (departure time endogenous). Each OD pair w has a fixed total demand D_w over a departure window $\mathcal{K}_w^{dep} \subseteq \mathcal{K}$ and a desired arrival time t_w^* . Departure time is an endogenous choice variable. Given EV penetration $p_w^{EV} \in [0, 1]$:

$$D_w^F = (1 - p_w^{EV}) D_w, \quad D_w^E = p_w^{EV} D_w. \quad (4)$$

The departure profile $d_{w,k} = \sum_r h_{w,r,k}^F + \sum_\omega h_{w,\omega,k}^E$ is an equilibrium *output*, not a given. This is the full DUE specification; the current CTM-DUE implementation supports this mode for EV demands with non-empty `departure_window_steps`.

Fixed-departure mode (departure time exogenous). When an exogenous departure profile $d_{w,k}$ is supplied, the model reduces to a fixed-departure dynamic assignment: equilibrium is imposed only across routes or activity plans *within* each departure step k . In this case the demand split is

$$d_{w,k}^F = (1 - p_{w,k}^{EV}) d_{w,k}, \quad d_{w,k}^E = p_{w,k}^{EV} d_{w,k}. \quad (5)$$

This is a restricted special case of the full DUE; benchmark tests in §1.15 use fixed-departure mode for comparability with published results.

1.2.4 Route and activity-plan sets

The ICV route set for OD w is $\mathcal{R}_w$. An EV *activity plan* $\omega = (r, s, \chi)$ combines a road path r , a *single* optional charging stop $s \in \mathcal{S} \cup \{\emptyset\}$, and a charging/V2G strategy χ (charging power profile, charging energy, discharging energy, dwell duration). The current implementation supports single-stop plans. Multi-stop plans (secondary charging after V2G discharge) are a planned rolling-horizon extension; until then, secondary charging after V2G must be handled by including a separate plan with the net energy already accounting for the discharge. The full activity-plan set for OD w at step k is $\Omega_{w,k}$. ICVs choose a route; EVs choose an activity plan.

Remark 2 (Activity-plan generation: theory vs. implementation). *The description below gives the full mathematical specification. In the current implementation, $\Omega_{w,k}$ is not generated automatically. The calling code provides pre-specified route alternatives and charging-stop templates directly via `EVDemand::candidate_route_indices` and `RouteAlternative::charging_stops` (`types.hpp`). Automatic generation of $\Omega_{w,k}$ by K -shortest-path enumeration, station reachability, and charging-template synthesis is a planned extension and not yet implemented in `evpt_internal.hpp`.*

Formally, $\Omega_{w,k}$ is defined as

$$\Omega_{w,k} = \{(r, s, \chi) : r \in \mathcal{R}_w^K, s \in \mathcal{S}_r \cup \{\emptyset\}, \chi \in \Xi_{s,k}, e_\ell - E_\ell^{rem} \geq e^{res}, \forall \ell\}. \quad (6)$$

1.2.5 Charging stations

The station set $\mathcal{S}$ collects service nodes embedded in the road network. Station s is characterised by plug count $\bar{B}_s$, maximum charging power $\bar{p}_s^{ch}$, maximum discharging power $\bar{p}_s^{dis}$, queue capacity $\bar{Q}_s$, dwell capacity $\bar{H}_s$, and grid-bus mapping $\text{bus}(s)$.

1.3 Modelling Assumptions

- A1 (LWR kinematic wave).** Road links follow first-order traffic flow theory (LWR) with triangular FD (2).
- A2 (FIFO).** Vehicles on the same link satisfy first-in-first-out: if $k_{i,a}^{in} \leq k_{j,a}^{in}$ then $k_{i,a}^{out} \leq k_{j,a}^{out}$.
- A3 (Equal PCE).** ICVs and EVs have identical passenger-car equivalence: $\text{PCE}^F = \text{PCE}^E = 1$. Class-specific PCE may be added as an extension.
- A4 (CTM/LTM for propagation).** CTM and LTM handle traffic propagation only; vehicle route choice and station activity are determined by the vehicle behaviour layer.
- A5 (ICV minimum-time).** The base ICV objective is minimum travel time. Fuel costs may be included in the generalised cost as an extension.
- A6 (EV economic optimum).** EVs minimise a generalised cost that includes travel time, station waiting, charging service time, energy price, battery degradation, and SOC feasibility (Section 1.5).
- A7 (Fleet representation).** Vehicles sharing the same OD, departure step, and behaviour type are aggregated into a fleet flow $x_{w,r,k}^F$ or $x_{w,\omega,k}^E$ (Section 1.6).
- A8 (Discrete SOC dynamics).** EV battery energy follows the discrete update (15).
- A9 (Mobility feasibility).** At every step: $e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}$.
- A10 (Sequential charge/discharge).** Charging and discharging are sequential phases of the station visit. An activity plan ω specifies either a charging session ($E_\omega^{ch} \geq 0, E_\omega^{dis} = 0$), a discharging (V2G) session ($E_\omega^{ch} = 0, E_\omega^{dis} \geq 0$), or both as consecutive phases. As shown in Proposition 1 below, any plan with simultaneously positive $p_{v,k}^{ch} > 0$ and $p_{v,k}^{dis} > 0$ at the same step k is strictly dominated in cost under the no-arbitrage price condition (26), so no binary mode variables are required in the typical operating regime.
- A11 (Station as service node).** EVs traverse arrival $\rightarrow$ queue $\rightarrow$ service $\rightarrow$ dwell $\rightarrow$ egress inside the network loading.
- A12 (Plug capacity).** $B_{s,k} \leq \bar{B}_s$ at every step.

1.4 Microscopic Vehicle Model

1.4.1 ICV microscopic state and route choice

ICV v has state $x_{v,k}^F = (l_{v,k}, p_v)$ comprising its current location and chosen path. For OD w departing at step k , the route-choice problem is

$$r_{v,k}^* = \arg \min_{r \in \mathcal{R}_w} T_{r,k}, \quad (7)$$

where the path travel time is computed by dynamic recursion along path $r = (a_1, a_2, \dots, a_m)$. This equation gives individual route choice in *fixed-departure mode* where departure step k is exogenous. In full DUE, ICVs jointly choose departure step and route to minimise $\Psi_{w,r,k}^F$ (which includes schedule delay; see §1.11.2).

$$k_{j+1} = k_j + \left\lceil \frac{T_{a_j, k_j}}{\Delta t} \right\rceil, \quad T_{r,k} = \sum_{j=1}^m T_{a_j, k_j}. \quad (8)$$

Under Wardrop time-equilibrium in *fixed-departure mode* the ICV travel-time condition is

$$T_{r,k} \begin{cases} = T_{w,k}^{\min} & \text{if } x_{w,r,k}^F > 0, \\ \geq T_{w,k}^{\min} & \text{if } x_{w,r,k}^F = 0. \end{cases} \quad (9)$$

Here $x_{w,r,k}^F = h_{w,r,k}^F$ is the equilibrium route flow at fixed departure step k . The generalised-cost condition for *full DUE* (with departure-time endogeneity and schedule delay) is given by (81)–(82) in §1.11.

Fuel consumption and emissions. Link-level fuel consumption for ICV v on link a is

$$F_{v,a,k} = L_a \left(\alpha_F + \frac{\beta_F}{\bar{v}_{a,k}} \right), \quad (10)$$

where α_F (L/km) is the distance-proportional coefficient, β_F (L·h/km) the congestion-sensitive coefficient, and $\bar{v}_{a,k} = L_a/T_{a,k}$ the mean link speed. Tailpipe CO₂ emissions are $G_{v,a,k} = \varepsilon^F F_{v,a,k}$.

1.4.2 EV microscopic state

EV v has a four-dimensional state

$$x_{v,k}^E = (l_{v,k}, e_{v,k}, \sigma_{v,k}, p_v), \quad (11)$$

where $l_{v,k}$ is its location, $e_{v,k}$ its battery energy (kWh), $\sigma_{v,k}$ its behavioural state, and p_v its activity plan. The behavioural state belongs to

$$\sigma_{v,k} \in \{ \text{driving, accessing_station, queueing, charging, discharging, dwelling, egressing, completed} \}. \quad (12)$$

1.4.3 EV driving energy consumption

Link-level energy consumption is

$$E_{v,a}^{\text{drive}} = \xi_v L_a \quad (\text{distance-linear model}), \quad (13)$$

or, with speed dependence,

$$E_{v,a,k}^{\text{drive}} = g_v(L_a, T_{a,k}, \bar{v}_{a,k}). \quad (14)$$

The mean link speed $\bar{v}_{a,k} = L_a/T_{a,k}$ links congestion directly to energy consumption.

1.4.4 EV SOC dynamics

The battery energy evolves as

$$e_{v,k+1} = e_{v,k} - E_{v,k}^{drive} + \eta_v^{ch} p_{v,k}^{ch} \Delta t - \frac{p_{v,k}^{dis} \Delta t}{\eta_v^{dis}}, \quad (15)$$

subject to $e_v^{min} \leq e_{v,k} \leq e_v^{max}$ and the *mobility feasibility* constraint

$$e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}, \quad (16)$$

where $E_{v,k}^{rem}$ is the energy required from the current position to the trip destination and e_v^{res} is the reserve threshold. Equation (16) is the critical coupling between V2G dispatch and mobility: discharging reduces the feasible plan set and may necessitate a second charging stop.

1.4.5 EV charging trigger

An EV *must* seek a charging stop if the safety constraint (16) would be violated without one:

$$e_{v,k} - E_{v,k}^{rem} < e_v^{res}. \quad (17)$$

With economic optimisation the trigger extends to any plan that reduces total cost:

$$\min_{\omega \in \Omega_{v,k}^{stop}} C_{v,\omega,k}^E < C_{v, \text{no stop}, k}^E, \quad (18)$$

meaning an EV with adequate SOC may still enter a station if low energy prices or V2G revenue make a stop economically attractive.

1.5 EV Economic Cost Function

The total generalised cost of activity plan ω for OD w departing at step k is

$$C_{w,\omega,k}^E = C_{w,\omega,k}^{travel} + C_{w,\omega,k}^{wait} + C_{w,\omega,k}^{service} + C_{w,\omega,k}^{energy} + C_{w,\omega,k}^{deg} + C_{w,\omega,k}^{schedule} + \phi_{w,\omega,k}^{soc}. \quad (19)$$

Travel-time cost. $C_{w,\omega,k}^{travel} = \alpha_t T_{\omega,k}$, where α_t (\$/h) is the value of time.

Waiting-time cost. If ω includes station s , with k_s^{arr} the arrival step:

$$C_{w,\omega,k}^{wait} = \alpha_w W_{s,k_s^{arr}}, \quad (20)$$

where $W_{s,k}$ is the expected queue waiting time at station s at step k . This term is zero if $s = \emptyset$.

Service-time cost. Charging and discharging are sequential: a vehicle first charges for duration T_{ω}^{ch} , then (if V2G) discharges for duration T_{ω}^{dis} , occupying a plug for the total time $T_{\omega}^{service} = T_{\omega}^{ch} + T_{\omega}^{dis}$:

$$T_{\omega}^{ch} = \frac{E_{\omega}^{ch}}{\eta_v^{ch} \bar{p}_s^{ch}}, \quad T_{\omega}^{dis} = \frac{E_{\omega}^{dis}}{\bar{p}_s^{dis}}, \quad (21)$$

and $C_{w,\omega,k}^{service} = \alpha_s T_{\omega}^{service}$.

Remark 3 (Energy-side convention). *Throughout this section, $p_{v,k}^{ch}$ and $p_{v,k}^{dis}$ are grid-side power quantities. The battery receives $\eta_v^{ch} p_{v,k}^{ch} \Delta t$ energy per charging step and releases $p_{v,k}^{dis} \Delta t / \eta_v^{dis}$ per*

discharging step, consistent with the SOC update (15). The battery-side charged energy and grid-side discharged energy for an activity plan are

$$E_{\omega}^{ch} = \sum_{\ell \in \mathcal{K}_{\omega}^s} \eta^{ch} p_{\omega,\ell}^{ch} \Delta t, \quad E_{\omega}^{dis} = \sum_{\ell \in \mathcal{K}_{\omega}^s} p_{\omega,\ell}^{dis} \Delta t. \quad (22)$$

Charging payments are based on grid-side energy drawn ($\sum_{\ell} p_{\omega,\ell}^{ch} \Delta t = E_{\omega}^{ch} / \eta^{ch}$); V2G revenue is based on grid-side energy delivered (E_{ω}^{dis}). This convention is consistent with the service-time formula (21) and the energy-cost term (23).

Because discharging extends the total plug-occupancy time $T^{service}$, it competes directly with queueing vehicles and enters the cost function without any binary mode variable.

Energy cost. Let $\pi_{s,\ell}^{ch}$ and $\pi_{s,\ell}^{dis}$ be the per-kWh charging price and discharging compensation at station s during service step ℓ . The net energy cost is

$$C_{w,\omega,k}^{energy} = \sum_{\ell \in \mathcal{K}_{\omega}^s} (\pi_{s,\ell}^{ch} p_{\omega,\ell}^{ch} - \pi_{s,\ell}^{dis} p_{\omega,\ell}^{dis}) \Delta t, \quad (23)$$

where charging is a positive cost and V2G revenue is negative.

Battery degradation cost. A linear equivalent-throughput model gives $C_{w,\omega,k}^{deg} = c^{deg}(E_{\omega}^{ch} + E_{\omega}^{dis})$, where c^{deg} is interpreted as an equivalent cost per kWh of mixed battery/grid-side throughput. If degradation is evaluated strictly on the battery side, replace the discharging term by $E_{\omega}^{dis} / \eta^{dis}$: $C_{w,\omega,k}^{deg} = c^{deg,ch} E_{\omega}^{ch} + c^{deg,dis} E_{\omega}^{dis} / \eta^{dis}$. A cycle-depth model $f^{deg}(\text{DoD}_{\omega}, E_{\omega}^{throughput})$ may replace either form for higher fidelity.

Schedule delay cost. With desired arrival time t_w^* :

$$C_{w,\omega,k}^{schedule} = \gamma_e \max(0, t_w^* - t_{w,\omega,k}^{arr}) + \gamma_l \max(0, t_{w,\omega,k}^{arr} - t_w^*). \quad (24)$$

SOC infeasibility penalty.

$$\phi_{w,\omega,k}^{soc} = M \cdot \mathbf{1}[\exists \ell : e_{\omega,\ell} - E_{\omega,\ell}^{rem} < e^{res}], \quad (25)$$

where M is a large penalty. In optimisation formulations the mobility feasibility (16) is imposed as a hard constraint instead.

Proposition 1 (Sequential charge/discharge under no-arbitrage prices). *Assume the effective round-trip V2G compensation does not exceed the charging cost after round-trip efficiency and degradation:*

$$\pi_{s,k}^{dis} \eta^{ch} \eta^{dis} \leq \pi_{s,k}^{ch} + c^{deg}. \quad (26)$$

Then for any activity plan ω with $p_{\omega,k}^{ch} > 0$ and $p_{\omega,k}^{dis} > 0$ at the same step k , there exists a plan ω' with strictly lower cost: $C_{w,\omega',k}^E < C_{w,\omega,k}^E$.

Proof. Consider any step k at which $p_{\omega,k}^{ch} > 0$ and $p_{\omega,k}^{dis} > 0$. The net battery-energy increment induced by the original plan at this step is

$$\Delta e_k = \eta^{ch} p_{\omega,k}^{ch} \Delta t - \frac{p_{\omega,k}^{dis} \Delta t}{\eta^{dis}}. \quad (27)$$

Construct an alternative plan ω' that achieves the same net SOC increment Δe_k but avoids simultaneous charging and discharging. If $\Delta e_k \geq 0$, set

$$p_{\omega',k}^{ch} = \frac{\Delta e_k}{\eta^{ch} \Delta t}, \quad p_{\omega',k}^{dis} = 0. \quad (28)$$

If $\Delta e_k < 0$, set

$$p_{\omega',k}^{ch} = 0, \quad p_{\omega',k}^{dis} = -\frac{\Delta e_k \eta^{dis}}{\Delta t}. \quad (29)$$

In both cases one verifies directly that $\eta^{ch} p_{\omega',k}^{ch} \Delta t - p_{\omega',k}^{dis} \Delta t / \eta^{dis} = \Delta e_k$, so ω' preserves the same SOC trajectory at step k .

Because the original plan simultaneously charges and discharges, it contains an internal cycling component that strictly increases total energy throughput and plug-occupancy time relative to ω' . Therefore $C_{\omega'}^{service} < C_{\omega}^{service}$ and $C_{\omega'}^{deg} < C_{\omega}^{deg}$.

Under the no-arbitrage condition (26), the eliminated cycling component cannot produce a net cost saving after accounting for round-trip efficiency and degradation, so the energy-cost term does not increase. Therefore $C_{w,\omega',k}^E < C_{w,\omega,k}^E$, proving that no cost-minimising plan contains simultaneous charging and discharging at the same step. $\square$

Remark 4 (Price condition). *Proposition 1 requires the no-arbitrage condition (26): the effective round-trip V2G revenue $\pi_{s,k}^{dis} \eta^{ch} \eta^{dis}$ must not exceed the charging cost $\pi_{s,k}^{ch}$ plus degradation c^{deg} . When this condition fails (strong V2G premium), simultaneous charge/discharge may be cost-improving; binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ should then be added. Under the no-arbitrage condition, binary variables are not needed and the problem remains a pure LP.*

1.5.1 EV equilibrium condition (fixed-departure special case)

In fixed-departure mode, the EV Wardrop condition at a given step k is:

$$h_{w,\omega,k}^E > 0 \Rightarrow C_{w,\omega,k}^E = C_{w,k}^{E,\min} := \min_{\omega' \in \Omega_{w,k}} C_{w,\omega',k}^E. \quad (30)$$

This is the restriction of the full DUE condition (83)–(84) to a single departure step; the OD-level minimum μ_w^E reduces to the step-level minimum $C_{w,k}^{E,\min}$ when the departure profile is fixed.

1.6 Fleet Aggregation Model

1.6.1 ICV departure-rate variables and demand conservation

The ICV DUE variable is $h_{w,r,k}^F \geq 0$: the number of ICVs of OD w departing at step k and choosing route r . The *full-DUE* demand constraint sums across the departure window:

$$\sum_{k \in \mathcal{K}_w^{dep}} \sum_{r \in \mathcal{R}_w} h_{w,r,k}^F = D_w^F \quad \forall w. \quad (31)$$

In *fixed-departure mode* the per-step constraint replaces (31):

$$\sum_{r \in \mathcal{R}_w} h_{w,r,k}^F = d_{w,k}^F \quad \forall w, k. \quad (32)$$

The notation $x_{w,r,k}^F$ used in earlier sections (route-flow after equilibration) equals $h_{w,r,k}^F$ at the fixed point.

1.6.2 EV activity-plan departure-rate variables and demand conservation

The EV DUE variable is $h_{w,\omega,k}^E \geq 0$: the number of EVs of OD w departing at step k and choosing activity plan ω . The *full-DUE* demand constraint is:

$$\sum_{k \in \mathcal{K}_w^{dep}} \sum_{\omega \in \Omega_{w,k}} h_{w,\omega,k}^E = D_w^E \quad \forall w. \quad (33)$$

In *fixed-departure mode*:

$$\sum_{\omega \in \Omega_{w,k}} h_{w,\omega,k}^E = d_{w,k}^E \quad \forall w, k. \quad (34)$$

1.6.3 Fleet-to-link inflow mapping

Total link inflow $u_{a,k}$ is the sum of contributions from ICV routes, EV activity plans, and station egress flows:

$$u_{a,k} = u_{a,k}^F + u_{a,k}^E + u_{a,k}^{egress}, \quad (35)$$

where dynamic-incidence indicators $\delta_{a,w,r,k}^F$ and $\delta_{a,w,\omega,k}^E$ specify whether a given fleet flow enters link a at step k ; these are determined endogenously by the CTM/LTM propagation layer.

1.6.4 Fleet-to-station arrival and departure

Station s arrival flow:

$$u_{s,k}^S = \sum_w \sum_{\substack{\omega \in \Omega_{w,k} \\ s \in \omega}} \delta_{s,w,\omega,k}^{arr} x_{w,\omega,k}^E, \quad (36)$$

and station departure flow:

$$o_{s,k} = \sum_w \sum_{\substack{\omega \in \Omega_{w,k} \\ s \in \omega}} \delta_{s,w,\omega,k}^{dep} x_{w,\omega,k}^E. \quad (37)$$

The departure flow $o_{s,k}$ re-enters the road network as $u_{a,k}^{egress}$ on the station's egress link.

1.6.5 Class-specific cumulative curves

Class-specific cumulative entry and exit counts sum to the aggregate:

$$N_{a,k}^{in} = \sum_{c \in \mathcal{C}} N_{a,k}^{in,c}, \quad N_{a,k}^{out} = \sum_{c \in \mathcal{C}} N_{a,k}^{out,c}. \quad (38)$$

Link occupancy at step k : $n_{a,k} = N_{a,k}^{in} - N_{a,k}^{out}$.

1.7 CTM-Based System-Level Network Loading

The *Cell Transmission Model* (CTM, Daganzo 1994) discretises each link into a sequence of cells, resolving within-link shockwave dynamics and supporting route-commodity tracking for DUE. The implementation resides in `src/ev_power_traffic/ctm_propagation.cpp`.

1.7.1 Cell discretisation and CFL condition

Link a is partitioned into M_a cells of equal length

$$\delta_a = \frac{L_a}{M_a}, \quad M_a = \max\left(1, \left\lfloor \frac{L_a}{v_a^{ff} \Delta t} \right\rfloor\right). \quad (39)$$

When $L_a \geq v_a^{ff} \Delta t$, the choice in (39) yields $\delta_a = L_a/M_a \geq v_a^{ff} \Delta t$, so the **CFL condition** $\delta_a \geq v_a^{ff} \Delta t$ is satisfied by construction. If $L_a < v_a^{ff} \Delta t$, no cell partition with $M_a \geq 1$ can satisfy this condition; the global time step must be reduced or sub-stepping must be used before applying the CTM. For the full network:

$$\Delta t \leq \min_{a \in \mathcal{A}^R} \frac{\delta_a}{v_a^{ff}}. \quad (40)$$

The jam occupancy per cell is

$$N_{a,m}^{jam} = k_a^j \delta_a = \frac{\bar{q}_a(v_a^{ff} + w_a)}{v_a^{ff} w_a} \delta_a. \quad (41)$$

1.7.2 Cell sending and receiving functions

$$S_{a,m,k} = \min\left(\frac{v_a^{ff}}{\delta_a} n_{a,m,k}, \bar{q}_a\right) \Delta t, \quad (42)$$

$$R_{a,m,k} = \min\left(\bar{q}_a, \frac{w_a}{\delta_a} (N_{a,m}^{jam} - n_{a,m,k})\right) \Delta t. \quad (43)$$

1.7.3 Inter-cell flow and occupancy update

The flow from cell m to cell $m + 1$:

$$y_{a,m \rightarrow m+1,k} = \min(S_{a,m,k}, R_{a,m+1,k}), \quad (44)$$

and the cell occupancy update:

$$n_{a,m,k+1} = n_{a,m,k} + y_{a,m-1 \rightarrow m,k} - y_{a,m \rightarrow m+1,k}. \quad (45)$$

1.7.4 Route-commodity tracking

For multi-route networks a per-route fraction is maintained:

$$\phi_{a,m,k}^r = \frac{n_{a,m,k}^r}{n_{a,m,k} + \varepsilon}, \quad n_{a,m,k} = \sum_r n_{a,m,k}^r, \quad (46)$$

and the per-route inter-cell transfer is $y_{a,m \rightarrow m+1,k}^r = \phi_{a,m,k}^r y_{a,m \rightarrow m+1,k}$.

Remark 5 (Proportional mixing approximation). Equation (46) propagates route identities by proportional mixing: each route receives a share of the inter-cell flow proportional to its current fraction of cell occupancy. This preserves aggregate link conservation but does not enforce route-level FIFO ordering. Vehicles from later-departing routes may be assigned to earlier exit slots. Exact route-specific FIFO tracking requires per-commodity cumulative curves and is handled by the LTM layer (Section 1.8).

1.7.5 Inter-link node model

At node i , the inter-link flow from link a to link b is

$$y_{a \rightarrow b,k} = \beta_{a,b,k} \min(S_{a,k}^{exit}, R_{b,k}^{entry}), \quad (47)$$

where $S_{a,k}^{exit} = S_{a,M_a-1,k}$, $R_{b,k}^{entry} = R_{b,0,k}$, and $\beta_{a,b,k}$ is the turning fraction derived from route fractions or given as exogenous input.

Remark 6 (Node model limitation). Equation (47) is a single-input/single-output approximation valid when each junction has at most one upstream link. For general merge nodes (multiple incoming links sharing one outgoing link), the individual terms $\beta_{a,b,k} \min(S_{a,k}^{exit}, R_{b,k}^{entry})$ may sum to exceed $R_{b,k}^{entry}$, violating the downstream receiving constraint. A proper node model maximises joint throughput subject to sending and receiving constraints simultaneously (see equations (54)–(55) in the LTM section), applying supply-side priority rules at diverge nodes.

Remark 7 (Current CTM implementation scope). The current `ctm_propagation.cpp` implementation applies equation (47) with precomputed turning fractions. Support for general diverge/merge node solvers (simultaneous multi-link capacity allocation) is a planned extension and is not yet enabled.

1.7.6 CTM-based Dynamic User Equilibrium (DUE)

The CTM is embedded in a DUE outer loop. Two solution methods are available:

MSA (heuristic baseline).

$$x^{(i+1)} = x^{(i)} + \frac{1}{i+1} (\hat{x}^{(i)} - x^{(i)}), \quad (48)$$

where $\hat{x}^{(i)}$ is the all-or-nothing assignment to shortest-time routes given CTM travel times from iteration i . The relative gap

$$\epsilon_i = \frac{\sum_r (T^r - T^{\min}) x_r^{(i)}}{\sum_r T^{\min} x_r^{(i)}} \quad (49)$$

drives convergence.

LP-based system-optimal formulation. Section 1.17.1 gives the SO-CTM LP that replaces MSA when system-optimal objectives are acceptable and optimality certificates are required.

Remark 8 (CTM vs LTM granularity). *CTM has $O(M_a|\mathcal{A}|T)$ state variables per simulation; LTM has $O(|\mathcal{A}|T)$. Table 1 in §1.15.7 gives the full trade-off.*

1.8 LTM-Based Meso-Micro Network Loading

The *Link Transmission Model* (LTM, Yperman 2007) propagates traffic using cumulative entry and exit curves rather than per-cell densities. It is the primary propagation layer for EV meso-micro coupling because it provides exact FIFO trajectory reconstruction at $O(|\mathcal{A}|T)$ cost.

1.8.1 Cumulative curves and occupancy

For link a , define cumulative entry and exit counts:

$$u_{a,k} = N_{a,k+1}^{in} - N_{a,k}^{in}, \quad v_{a,k} = N_{a,k+1}^{out} - N_{a,k}^{out}. \quad (50)$$

Link occupancy: $n_{a,k} = N_{a,k}^{in} - N_{a,k}^{out}$.

1.8.2 Free-flow and backward-wave delays

$$\tau_a^{ff} = \left\lceil \frac{L_a}{v_a^{ff} \Delta t} \right\rceil, \quad \tau_a^{bw} = \left\lceil \frac{L_a}{w_a \Delta t} \right\rceil. \quad (51)$$

Remark 9 (Initial history curves). *The sending and receiving functions (52)–(53) reference $N_{a,k-\tau_a^{ff}}^{in}$ and $N_{a,k-\tau_a^{bw}}^{out}$ for $k < \tau$. For an empty initial network: $N_{a,k}^{in} = N_{a,k}^{out} = 0$ for all $k \leq 0$. Non-zero pre-history (e.g., background demand already en route) must be supplied externally before the first simulation step.*

1.8.3 LTM sending and receiving functions

$$S_{a,k} = \min \left(N_{a,k-\tau_a^{ff}}^{in} - N_{a,k}^{out}, \bar{q}_a \Delta t \right), \quad (52)$$

$$R_{a,k} = \min \left(N_{a,k-\tau_a^{bw}}^{out} + N_a^{jam} - N_{a,k}^{in}, \bar{q}_a \Delta t \right). \quad (53)$$

The boundary flow constraints are $0 \leq v_{a,k} \leq S_{a,k}$ and $0 \leq u_{a,k} \leq R_{a,k}$.

Remark 10 (Discrete-offset convention). *In the current implementation, entry flows at step k are written to $N_{a,k+1}^{in}$ (flows are increments between steps), and exit flows are written after the sending calculation. Depending on the update ordering, the first observable exit index in the code may appear at $k + \tau_a^{ff} + 1$ or $k + \tau_a^{ff} + 2$, rather than the theoretical $k + \tau_a^{ff}$. This is an implementation-indexing convention arising from the entry-increment and outflow write-back ordering, not an intrinsic LTM property. The theoretical minimum travel time remains $\tau_a^{ff} \Delta t$.*

1.8.4 LTM node model

For intersection node i with incoming link set $\mathcal{A}^{in}(i)$ and outgoing link set $\mathcal{A}^{out}(i)$:

$$\sum_{b \in \mathcal{A}^{out}(i)} y_{a,b,k} \leq S_{a,k}, \quad a \in \mathcal{A}^{in}(i), \quad (54)$$

$$\sum_{a \in \mathcal{A}^{in}(i)} y_{a,b,k} \leq R_{b,k}, \quad b \in \mathcal{A}^{out}(i). \quad (55)$$

Link flows satisfy

$$v_{a,k} = \sum_{b \in \mathcal{A}^{out}(i)} y_{a,b,k}, \quad (56)$$

$$u_{b,k} = \sum_{a \in \mathcal{A}^{in}(i)} y_{a,b,k} + d_{b,k}^{orig} + o_{s \rightarrow b,k}, \quad (57)$$

where $d_{b,k}^{orig}$ is exogenous origin demand and $o_{s \rightarrow b,k}$ is the egress flow entering link b from station s .

1.8.5 FIFO-consistent vehicle trajectory reconstruction

The LTM layer is mesoscopic and does not explicitly store individual trajectories. They are recovered by FIFO-consistent cumulative-count inversion.

For a vehicle entering link a during step k , its FIFO rank lies in the cumulative interval

$$\xi_{v,a} \in (N_{a,k}^{in}, N_{a,k+1}^{in}]. \quad (58)$$

If vehicles are represented individually, the j -th vehicle entering during step k is assigned $\xi_{v,a} = N_{a,k}^{in} + j$; for continuous flows a representative rank within the interval is used. The exit step is then

$$k_{v,a}^{out} = \min\{k : N_{a,k}^{out} \geq \xi_{v,a}\}. \quad (59)$$

Repeating over the activity path yields the full trajectory: $\mathcal{T}_v = \{(a, k_{v,a}^{in}, k_{v,a}^{out}) : a \in p_v\}$.

Station ranks are assigned analogously in cumulative intervals, where $U_{s,k}$ is cumulative arrivals, $B_{s,k}^{cum}$ cumulative service starts, $C_{s,k}^{cum}$ cumulative completions, and $O_{s,k}$ cumulative departures at station s by step k : arrivals during step k occupy $(U_{s,k}, U_{s,k+1}]$, service starts occupy $(B_{s,k}^{cum}, B_{s,k+1}^{cum}]$, and completions occupy $(C_{s,k}^{cum}, C_{s,k+1}^{cum}]$. For the vehicle arriving as the $\xi_{v,s}^{arr}$ -th at station s :

$$k_{v,s}^{start} = \min\{k : B_{s,k}^{cum} \geq \xi_{v,s}^{arr}\}, \quad (60)$$

$$k_{v,s}^{end} = \min\{k : C_{s,k}^{cum} \geq \xi_{v,s}^{start}\}, \quad (61)$$

$$k_{v,s}^{dep} = \min\{k : O_{s,k} \geq \xi_{v,s}^{end}\}. \quad (62)$$

This links aggregate LTM states to individual EV records without a full microscopic simulation.

1.9 Charging Stations as Endogenous Service Nodes

1.9.1 Station state and dynamics

Station s tracks three aggregate populations: $Q_{s,k}$ (queueing), $B_{s,k}$ (in service), $H_{s,k}$ (post-service dwell). The state transitions are

$$Q_{s,k+1} = Q_{s,k} + u_{s,k}^S - b_{s,k}, \quad (63)$$

$$B_{s,k+1} = B_{s,k} + b_{s,k} - c_{s,k}, \quad (64)$$

$$H_{s,k+1} = H_{s,k} + c_{s,k} - o_{s,k}. \quad (65)$$

1.9.2 Service admission

$$b_{s,k} \leq Q_{s,k} + u_{s,k}^S, \quad B_{s,k} + b_{s,k} \leq \bar{B}_s. \quad (66)$$

1.9.3 Service completion

$$c_{s,k} \leq B_{s,k}, \quad c_{s,k} \leq \mu_{s,k}^{step}, \quad (67)$$

where in the fluid station model

$$\mu_{s,k}^{step} = \frac{\bar{B}_s \Delta t}{T_{s,k}^{service}} \quad (68)$$

is the per-step service completion capacity (veh/step). In the individual-vehicle mode, $c_{s,k} = \sum_v \mathbf{1}[k_{v,s}^{end} = k]$.

1.9.4 Station departure

$$o_{s,k} \leq H_{s,k} + c_{s,k}, \quad o_{s,k} \leq R_{s,k}^{egress}, \quad (69)$$

where $R_{s,k}^{egress}$ is the receiving capacity of the downstream road link. An overcrowded downstream link prevents EV departure even after service completion.

1.9.5 Queue spillback to road

If $\bar{Q}_s < \infty$ is the queue capacity, the access-link receiving capacity is reduced as the queue fills:

$$R_{s,k}^{access} = R_{s,k}^0 \left(1 - \frac{Q_{s,k}}{\bar{Q}_s} \right)_+, \quad (70)$$

so that $R_{s,k}^{access} = 0$ when $Q_{s,k} \geq \bar{Q}_s$. This feedback propagates station congestion back into the road network.

Remark 11 (Spillback and egress feedback: partially wired). *Equations (69) and (70) specify the egress-capacity bound $R_{s,k}^{egress}$ and the access-link capacity reduction, respectively. In the current implementation, CTM applies a cumulative-arrival proxy, and LTM uses queue-based receiving reduction. Thus, station arrivals do feed back into upstream congestion. However, this is still a partial formulation relative to the full bidirectional theoretical model.*

1.10 Charging and V2G Feedback on Traffic

Charging and V2G decisions affect traffic through three channels.

Channel 1: Route and plan choice. Low energy prices or high V2G compensation may induce EVs to choose a longer route that passes a favoured station:

$$C_{\omega_B,k}^E < C_{\omega_A,k}^E \quad \Rightarrow \quad x_{w,\omega_B,k}^E > 0, \quad (71)$$

changing link inflows $u_{a,k}$ and thus congestion.

Channel 2: SOC depletion and secondary charging. V2G discharge reduces $e_{v,k}$, potentially triggering the safety condition (17) later in the trip. A secondary charging stop adds an extra station visit, increases $u_{s',k}^S$ at a downstream station s' , and extends total trip time.

Channel 3: Dwell time and egress. Charging duration $T_\omega^{ch} = E_\omega^{ch}/(\eta^{ch}\bar{p}_s^{ch})$ directly determines how long a vehicle occupies a plug. Larger charging energy increases $B_{s,k}$, reduces $\bar{B}_s - B_{s,k}$, and can grow the queue $Q_{s,k}$, which eventually activates the spillback mechanism (70).

Summary of coupling.

$$\underbrace{(E^{ch}, E^{dis}, \pi^{ch}, \pi^{dis})}_{\text{charging/V2G strategy}} \longrightarrow \underbrace{(u_{s,k}^S, o_{s,k}, u_{a,k})}_{\text{station and link flows}} \longrightarrow \underbrace{(N^{in}, N^{out}, T_{a,k})}_{\text{LTM/CTM state}} \longrightarrow \underbrace{C_{w,\omega,k}^E}_{\text{EV cost feedback}}. \quad (72)$$

1.11 Dynamic Equilibrium Model

This module supports two assignment modes: (i) **Full DUE**: departure time, route, and EV activity plan are endogenous choices under OD total-demand constraints (31)–(33); equilibrium costs are compared across all departure steps and alternatives. (ii) **Fixed-departure DUE**: the departure profile $d_{w,k}$ is given; only route/activity-plan choice is equilibrated within each departure step under constraints (32)–(34).

Remark 12 (Implementation scope: Full DUE for EV windows). *The CTM-DUE solver supports both modes. With `DUEOptions::due_mode=DUEMode::FullEndogenous`, each EV demand with non-empty `EVDemand::departure_window_steps` is assigned over the full (k, ω) window: the all-or-nothing direction minimises generalized cost across departure step, route/activity plan, charging cost, V2G credit, and schedule-delay penalties. The result records both route totals and disaggregated `flow_by_demand_departure_route`. Fixed-departure mode remains the default for backward compatibility. Multi-class ICV departure-time endogeneity is still limited to fixed `ICVDemand::departure_step`; extending the ICV multi-class branch to cross-step departure windows is a remaining implementation extension.*

Unless stated otherwise, the mathematical formulations in this section describe full DUE (i); benchmark cases explicitly state whether they use full-endogenous or fixed-departure mode.

1.11.1 Dynamic Network Loading operator

The DNL operator maps departure-rate flows to network states:

$$\mathcal{D} : (h^F, h^E) \mapsto (T, W, Q, o, N^{in}, N^{out}). \quad (73)$$

1.11.2 Experienced-cost operators

After DNL produces network states, the *experienced-cost operators* convert states to generalised path costs:

$$\Psi_{w,r,k}^F(h) := \alpha_t T_{w,r,k}^{exp}(h) + C_{w,r,k}^{schedule,F}(h), \quad (74)$$

$$\Psi_{w,\omega,k}^E(h) := C_{w,\omega,k}^E(h), \quad (75)$$

where $T_{w,r,k}^{exp}(h)$ is the FIFO-reconstructed experienced travel time of a departure- k vehicle on route r , $C_{w,\omega,k}^E$ is the EV generalised cost from (??), and the ICV schedule-delay penalty is

$$C_{w,r,k}^{schedule,F}(h) = \gamma_e \max(0, t_w^* - t_{w,r,k}^{arr}) + \gamma_l \max(0, t_{w,r,k}^{arr} - t_w^*), \quad (76)$$

with $t_{w,r,k}^{arr} = k\Delta t + T_{w,r,k}^{exp}$, desired arrival time t_w^* , and early/late penalties $\gamma_e, \gamma_l \geq 0$. Setting $\gamma_e = \gamma_l = 0$ recovers the travel-time-only special case.

1.11.3 ICV and EV choice operators

The DUE minimum costs are OD-level minima across *all* departure steps and alternatives in the departure window:

$$\mu_w^F := \min_{k \in \mathcal{K}_w^{dep}, r \in \mathcal{R}_w} \Psi_{w,r,k}^F(h), \quad (77)$$

$$\mu_w^E := \min_{k \in \mathcal{K}_w^{dep}, \omega \in \Omega_{w,k}} \Psi_{w,\omega,k}^E(h). \quad (78)$$

The ICV assignment operator $\mathcal{A}^F$ and EV assignment operator $\mathcal{A}^E$ map experienced costs to departure-rate flows. Under all-or-nothing (AON) assignment for full DUE, the entire OD demand is assigned to the minimiser across all (k, r) (resp. (k, ω)):

$$\hat{h}_{w,r,k}^F = \begin{cases} D_w^F & (r, k) \in \arg \min_{r', k'} \Psi_{w,r',k'}^F, \\ 0 & \text{otherwise,} \end{cases} \quad (79)$$

$$\hat{h}_{w,\omega,k}^E = \begin{cases} D_w^E & (\omega, k) \in \arg \min_{\omega', k'} \Psi_{w,\omega',k'}^E, \\ 0 & \text{otherwise.} \end{cases} \quad (80)$$

Ties are broken by uniform splitting. A logit stochastic model gives $P_{w,\omega,k}^E \propto \exp(-\theta \Psi_{w,\omega,k}^E)$ and is normalised so that $\sum_{k,\omega} h_{w,\omega,k}^E = D_w^E$.

In *fixed-departure mode*, the minimisation in (77)–(80) is restricted to the given departure step k , and D_w^F, D_w^E are replaced by $d_{w,k}^F, d_{w,k}^E$.

1.11.4 Wardrop DUE conditions

At a full DUE h^* , no vehicle can reduce its experienced cost by unilaterally changing departure time, route, or activity plan:

$$h_{w,r,k}^{F,*} > 0 \Rightarrow \Psi_{w,r,k}^F(h^*) = \mu_w^{F,*}, \quad (81)$$

$$h_{w,r,k}^{F,*} = 0 \Rightarrow \Psi_{w,r,k}^F(h^*) \geq \mu_w^{F,*}, \quad (82)$$

$$h_{w,\omega,k}^{E,*} > 0 \Rightarrow \Psi_{w,\omega,k}^E(h^*) = \mu_w^{E,*}, \quad (83)$$

$$h_{w,\omega,k}^{E,*} = 0 \Rightarrow \Psi_{w,\omega,k}^E(h^*) \geq \mu_w^{E,*}. \quad (84)$$

Here $\mu_w^{F,*}$ and $\mu_w^{E,*}$ are the equilibrium (minimised) OD costs from (77)–(78) evaluated at h^* .

Remark 13 (Relationship to fixed-departure condition). *The condition in (30) is the fixed-departure special case of (83)–(84) with the minimum taken only over ω at a given k .*

1.11.5 Fixed-point equilibrium

The coupled system is at equilibrium when

$$(h^{F,*}, h^{E,*}) = \mathcal{A}(\mathcal{D}(h^{F,*}, h^{E,*})), \quad (85)$$

i.e. given the congestion and queues produced by the current departure rates, no vehicle has an incentive to change its departure time, route, or activity plan.

1.12 Solution Algorithm: MSA for Mixed Fleet DUE

Starting from initial flows $h^{F,(0)}, h^{E,(0)}$ satisfying the demand constraints, each iteration i proceeds as:

1. **DNL forward pass:** $(T^{(i)}, W^{(i)}, Q^{(i)}, o^{(i)}) = \mathcal{D}(h^{F,(i)}, h^{E,(i)})$.
2. **Compute experienced costs:** Evaluate $\Psi_{w,r,k}^{F,(i)}$ and $\Psi_{w,\omega,k}^{E,(i)}$ via (74)–(75) for all (w, r, k) and (w, ω, k) .
3. **AON assignment (cross- k):** For full DUE, assign each OD's total demand to the single minimiser across all departure steps and alternatives:

$$\hat{h}_{w,r,k}^{F,(i)} = \begin{cases} D_w^F & (r, k) \in \arg \min_{r', k'} \Psi_{w,r',k'}^{F,(i)}, \\ 0 & \text{otherwise,} \end{cases} \quad (86)$$

and analogously for $\hat{h}_{w,\omega,k}^{E,(i)}$ with D_w^E . In fixed-departure mode, the minimisation is restricted to each k with demand $d_{w,k}^F$ (resp. $d_{w,k}^E$).

4. **MSA update** (step size $\lambda_i = 1/(i+1)$):

$$h^{F,(i+1)} = h^{F,(i)} + \lambda_i (\hat{h}^{F,(i)} - h^{F,(i)}), \quad h^{E,(i+1)} = h^{E,(i)} + \lambda_i (\hat{h}^{E,(i)} - h^{E,(i)}). \quad (87)$$

5. **Convergence check.** The *ICV relative gap* (OD-level minimum):

$$\epsilon_i^F = \frac{\sum_{w,k,r} h_{w,r,k}^{F,(i)} (\Psi_{w,r,k}^{F,(i)} - \mu_w^{F,(i)})}{\sum_w D_w^F \mu_w^{F,(i)}}, \quad (88)$$

where $\mu_w^{F,(i)} = \min_{k,r} \Psi_{w,r,k}^{F,(i)}$. The *EV relative gap*:

$$\epsilon_i^E = \frac{\sum_{w,k,\omega} h_{w,\omega,k}^{E,(i)} (\Psi_{w,\omega,k}^{E,(i)} - \mu_w^{E,(i)})}{\sum_w D_w^E |\mu_w^{E,(i)}| + \varepsilon}, \quad (89)$$

where $\mu_w^{E,(i)} = \min_{k,\omega} \Psi_{w,\omega,k}^{E,(i)}$ and $\varepsilon > 0$ guards against a near-zero denominator when EV costs include V2G revenue that can make $\mu_w^{E,(i)}$ negative. Stop when $\max(\epsilon_i^F, \epsilon_i^E) < \epsilon^{tol}$.

1.13 Key Output Metrics

Traffic metrics. Total travel time in network: $TTT = \sum_{a,k} n_{a,k} \Delta t$. Total system travel time: $TSTT = \sum_{w,r,k} x_{w,r,k}^F T_{w,r,k} + \sum_{w,\omega,k} x_{w,\omega,k}^E T_{w,\omega,k}$. Network throughput: $\sum_{a \in \mathcal{A}^{sink}} N_{a,T}^{out}$.

Station metrics. Maximum queue $Q_s^{max} = \max_k Q_{s,k}$, average waiting time $\bar{W}_s = \sum_k Q_{s,k} \Delta t / \sum_k u_{s,k}^S$, and plug utilisation $\rho_s = \sum_k B_{s,k} / (\bar{B}_s T)$. The step-level expected waiting time used in the EV cost (20) is estimated by the fluid approximation $W_{s,k} \approx Q_{s,k} / \mu_{s,k}^{hour}$, where $\mu_{s,k}^{hour} = \bar{B}_s / \bar{T}_{s,k}^{service}$ (measured in veh/hour when $\bar{T}_{s,k}^{service}$ is in hours) gives $W_{s,k}$ in hours. The equivalent per-step capacity is $\mu_{s,k}^{step} = \mu_{s,k}^{hour} \Delta t = \bar{B}_s \Delta t / \bar{T}_{s,k}^{service}$ (veh/step).

EV energy metrics. Total charging energy $E_{tot}^{ch} = \sum_{s,k} P_{s,k}^{ch} \Delta t$, total V2G energy $E_{tot}^{dis} = \sum_{s,k} P_{s,k}^{dis} \Delta t$, net load energy $E_{tot}^{net} = E_{tot}^{ch} - E_{tot}^{dis}$, and V2G potential:

$$P_{s,k}^{v2g,pot} = \sum_{v \in \mathcal{V}_{s,k}^{dwell}} \min \left(\bar{p}_v^{dis}, \frac{e_{v,k} - e_v^{res}}{\Delta t} \eta_v^{dis} \right). \quad (90)$$

Equation (90) gives the *one-step* V2G power potential for vehicles currently dwelling at the station. Multi-period V2G dispatch additionally requires intertemporal constraints on SOC (15) and remaining dwell time $k_{v,s}^{dep} - k$ to ensure feasibility across the full dwell window.

Station net load. Station charging and discharging powers are $P_{s,k}^{ch} = \sum_v p_{v,k}^{ch}$ and $P_{s,k}^{dis} = \sum_v p_{v,k}^{dis}$, giving net load $P_{s,k}^{net} = P_{s,k}^{ch} - P_{s,k}^{dis}$ and bus-level EV injection $P_{b,k}^{EV} = \sum_{s: \text{bus}(s)=b} P_{s,k}^{net}$.

1.14 Correctness Checklist

The following invariants must hold throughout every simulation:

1. **Vehicle conservation.** $N_{a,k}^{out} \leq N_{a,k}^{in}$ for all a, k .
2. **Road capacity.** Admitted link inflows and outflows satisfy $u_{a,k} \leq \bar{q}_a \Delta t$ and $v_{a,k} \leq \bar{q}_a \Delta t$. Exogenous origin demand may exceed this value; any excess is stored in an origin queue and admitted in subsequent steps.
3. **Jam storage.** $n_{a,k} \leq N_a^{jam}$.
4. **Plug capacity.** $B_{s,k} \leq \bar{B}_s$.
5. **SOC feasibility.** $e_v^{min} \leq e_{v,k} \leq e_v^{max}$ and $e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}$.
6. **FIFO.** If $k_{i,a}^{in} \leq k_{j,a}^{in}$ then $k_{i,a}^{out} \leq k_{j,a}^{out}$.
7. **Sequential charge/discharge** (Proposition 1). $p_{v,k}^{ch} \cdot p_{v,k}^{dis} = 0$ at every step k when the no-arbitrage condition (26) holds; sequential behaviour then follows from cost optimality alone, without binary mode variables. When V2G compensation substantially exceeds charging cost after round-trip efficiency, binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ should be added.

1.15 Numerical Results

The following results are produced by five test suites, all assertions passing:

- `test_ltm_stress.cpp` — 20 cases, 1532 assertions (LTM wave propagation, CFL, conservation).
- `test_ctm_ltm_comparison.cpp` — 5 cases, 118 assertions (CTM vs. LTM throughput and travel-time accuracy).
- `test_ev_power_traffic_report.cpp` — 4 cases, 11 assertions (fleet assignment and V2G simulation with per-step output).
- `test_ev_power_traffic_joint_opt_d.cpp` — 10 cases, 93 assertions (joint social-welfare optimizer, FormD).
- `test_ev_power_traffic_largescale.cpp` — 3 cases, 14 assertions (10,000-EV scalability benchmark on IEEE-118 power grid).

Results are organised in two groups. The first group (§§1.15.2–??) contains **implementation diagnostics**: tests that characterise known numerical artefacts (integer-delay, CFL) and compare CTM vs. LTM behaviour. These are diagnostic results for implementation quality, not validation of the DUE equilibrium model. The second group (§§1.15.8–1.15.15) contains **model validation tests**: fleet assignment, V2G optimality, and scalability results that exercise the DUE and joint-optimisation layers.

1.15.1 Implementation diagnostics

These tests characterise the discrete-time LTM and CTM numerical behaviour; they are not model-validation results for the DUE equilibrium. Production DUE runs should use time steps and link discretisations for which the artefacts quantified below (sustained-capacity degradation, travel-time overestimation at low CFL numbers) are controlled or verified by time-step refinement tests.

1.15.2 Single-link LTM: propagation delay and CFL

For a single urban arterial link ($L = 2$ km, $v^{ff} = 50$ km/h, $w = 16.7$ km/h, $\bar{q} = 900$ veh/h, $\Delta t = 2$ min):

$$v^{ff} \Delta t = 50 \times \frac{1}{30} = 1.67 \text{ km} < L = 2 \text{ km} \quad [\text{CFL satisfied}],$$

$$\tau^{ff} = \lceil \frac{2}{1.67} \rceil = 2, \quad \tau^{bw} = \lceil \frac{2}{0.556} \rceil = 4, \quad N^{jam} \approx 144 \text{ veh}, \quad \text{cap_step} = 30 \text{ veh}.$$

A 20-vehicle pulse (well below $\text{cap_step} = 30$) enters at step 0. The LTM (Group XXI) shows:

- $N^{in}[T] = 20$ (full pulse absorbed),
- $N^{out}[T] = 20$ (all vehicles drain; $T = 30 \gg \tau^{ff} + \tau^{bw} = 6$),
- cumulative curves are monotone non-decreasing for all k ,
- CTM gives identical conservation ($N^{in} = N^{out} = 20$).

1.15.3 Freeway sustained-capacity: LTM vs CTM throughput

Freeway link ($L = 10$ km, $v^{ff} = 100$ km/h, $w = 25$ km/h, $\bar{q} = 2000$ veh/h, $\Delta t = 5$ min):

$$v^{ff} \Delta t = 8.33 \text{ km} < L = 10 \text{ km} \quad [\text{CFL satisfied}],$$

$$\tau^{ff} = 2, \quad \tau^{bw} = 5, \quad N^{jam} = 1000 \text{ veh}, \quad \text{cap_step} = 166.7 \text{ veh}.$$

With $T = 80$ steps of sustained capacity demand (Group XXII):

Model	Exit vehicles	Fraction of $T \cdot \text{cap_step}$
CTM	13 300	99.7%
LTM	8 833	66.2%

The CTM nearly matches $T \cdot \text{cap_step} = 13\,333$; in the current single-cell implementation, the inject-propagate update ordering allows a vehicle entering at step k to exit at the same step (a zero-delay implementation artefact that occurs when too few cells are used). The LTM throughput is lower due to a discretisation-induced storage–delay mismatch: with $\tau^{ff} = 2$ and $\tau^{bw} = 5$, the discrete delay product gives $\bar{q} \Delta t (\tau^{ff} + \tau^{bw}) = 166.7 \times 7 = 1167$ veh, which exceeds $N^{jam} = 1000$ veh. This causes an oscillatory fill/drain pattern. A crude fill/block-cycle approximation gives throughput on the order of $6/(6+5) \approx 55\%$ of capacity; the finite-horizon implementation reports 66.2% because start-up transients, terminal drainage, and the

discrete update convention allow additional vehicles to exit before the horizon closes. The key point is not the exact fraction but the presence of a storage–delay mismatch under the coarse $\Delta t = 5$ min discretisation. **This is a numerical artefact** that should be controlled by reducing Δt , using the corrected discrete jam density $N_a^{jam, disc} = \bar{q}_a \Delta t (\tau_a^{ff} + \tau_a^{bw})$, or a corrected update convention — it is not an intrinsic LTM property.

1.15.4 CFL-violation pathology

Intentionally violating CFL by using $L = 0.5$ km with the same freeway parameters ($v^{ff} \Delta t = 8.33$ km $\gg L$, Group XXIII):

Model	Average throughput	Fraction of cap
LTM	150 veh/h	7.5% (collapses)
CTM coarse (same Δt)	2000 veh/h	100% (physically invalid)
CTM fine ($\Delta t_{ctm} = 0.2$ min, CFL OK)	1999 veh/h	100% (correct)

The LTM collapses because $\tau^{ff} = \tau^{bw} = 1$, making $N^{jam} = 50$ while $\text{cap_step} = 166.7$, a fundamental contradiction. The coarse CTM spikes cell occupancies above N^{jam} but drains them in the same step, giving full capacity at a physically meaningless state. The fine CTM with CFL-satisfying sub-steps recovers correct behaviour. **Remedy:** always enforce $L \geq v^{ff} \Delta t$ at the modelling stage.

1.15.5 Four-link arterial corridor

Four copies of the urban arterial link (Group XXIV, $T = 120$):

Model	Exit vehicles	Dead-band	Prediction
LTM	1992	12 steps ($= 4(\tau^{ff} + 1)$)	$(T - 12) \times 30 = 3240$
CTM	3576	3 steps ($= N_L - 1$)	$(T - 3) \times 30 = 3510$

The LTM dead-band is the sum of per-link discrete offsets; the CTM dead-band is one step per inter-link crossing (zero-delay per link). LTM exits 1992 vehicles $\approx 61\%$ of its ceiling (same discretisation-induced oscillatory throughput phenomenon as Group XXII; each link contributes a $\tau^{bw} = 4$ blocking cycle).

1.15.6 Travel-time accuracy: LTM offset vs CTM zero-delay

Group XXV tests a 12-vehicle pulse on both a short link ($L = 1$ km, $\Delta t = 1$ min) and a freeway link ($L = 10$ km, $\Delta t = 5$ min):

Link	True ff (min)	LTM 1st exit	CTM step-0 exit	LTM overestimate
Short	1.2	step 4 = 4 min	10.0 veh	+233%
Freeway	6.0	step 4 = 20 min	10.0 veh	+233%

Both models have $\tau^{ff} = 2$; the LTM discrete-offset convention adds two more steps ($\tau^{ff} + 2 = 4$), placing the true delay between the CTM zero-lower-bound and the LTM upper-bound.

Table 1: CTM vs LTM: structural and computational comparison.

Feature	CTM (Daganzo 1994)	LTM (Yperman 2007)
State variable	Cell occupancy $n_{a,m,k}$	Cumulative curves $N_{a,k}^{in}, N_{a,k}^{out}$
State size	$O(M_a \mathcal{A} T)$	$O(\mathcal{A} T)$
CFL	$\delta_a \geq v_a^{ff} \Delta t$ (explicit)	$\tau^{ff} = \lceil L/(v^{ff} \Delta t) \rceil$ (implicit)
Travel delay	Zero-delay artefact in single-cell implementation	$\tau^{ff} \Delta t$ in theory; current implementation adds 1–2 indexing steps
FIFO consistency	Approximate (cell-averaged)	Exact (cumulative-count inversion)
Backward wave	Cell-to-cell propagation	Exact τ^{bw} look-back on N^{out}
Oscillatory throughput	Not observed in tested CFL-satisfying multi-cell cases	Possible under coarse time steps when discrete storage and delay are mismatched
Route tracking	Per-cell commodity fractions	Per-class cumulative curves
LP formulation size	$O((\sum_a M_a + \mathcal{R}) T)$	$O(\mathcal{A} T)$
Individual EV coupling	Via commodity fraction reconstruction	Direct FIFO inversion of N^{out}
Typical use	System-level DUE, shockwave analysis	Meso-micro EV-charging coupling
Implementation	<code>ctm_propagation.cpp</code>	<code>ltm_propagation.cpp</code>

1.15.7 CTM vs LTM: model comparison

1.15.8 Model validation tests

1.15.9 EV penetration sweep

A single station ($\bar{B} = 10$ plugs, mean service $\bar{\mu}^{-1} = 0.50$ h) on a 1-link corridor; 40% of EVs require charging each step: Queue lengths grow linearly with penetration. At 70% the arrival

Table 2: EV penetration sweep: station loading.

EV penetration	Max queue (veh)	Avg plug utilisation
10%	449	60.3%
30%	1 936	60.6%
50%	3 426	60.6%
70%	4 915	60.6%

rate ($1500 \times 0.7 \times 0.4 = 420$ veh/h) exceeds service capacity ($10/0.5 = 20$ veh/h) by factor 21, confirming the runaway queue prediction.

1.15.10 Individual EV trajectories and V2G

Three EV agents traverse a 3-link corridor (6.5 km total, $T = 96$ steps, $\Delta t = 0.25$ h): No EV triggered the safety condition (17). For the most vulnerable agent (EV-low, $e_0 = 16$ kWh), the remaining energy after traversing the full 6.5 km corridor is $e_0 - E^{drive} = 16.0 - 6.5 \times 0.20 = 14.7$ kWh; since $14.7 > e^{res} = 8$ kWh, no charging stop is triggered. FIFO trajectory recon-

Table 3: Individual EV agent profiles. Battery = 80 kWh, consumption = 0.20 kWh/km, reserve = 8 kWh.

Agent	Initial SOC	Initial energy (kWh)	Final SOC
EV-high	85%	68.0	83.4%
EV-mid	45%	36.0	43.4%
EV-low	20%	16.0	18.4%

struction (59) verified for a 20-vehicle cohort: all 20 trajectories satisfy temporal consistency, $k^{out} \geq k^{in}$, minimum travel time $\geq \tau^{ff} \Delta t$, and strict FIFO order.

1.15.11 Fleet assignment: station-capacity benchmark (4-node, 5 EVs)

Network and parameters. A 4-node, 4-link network with two routes from node 1 to node 4 ($\Delta t = 1$ h, $T = 6$, default station price 0.30 \$/kWh). Five EVs each request 20 kWh of charging energy:

- **Route 1** (near): links 11→12; free-flow 0.50 h; station 101 at node 2 — 40 kW total, 2 fast plugs at 20 kW/veh.
- **Route 2** (far): links 21→22; free-flow 1.00 h; station 102 at node 3 — 60 kW total, 3 fast plugs at 20 kW/veh.

All 5 vehicles have the same origin/destination (node 1→4) departing at step 0. Table 4 compares three assignment algorithms.

Table 4: Fleet assignment comparison: 4-node benchmark, 5 EVs. h_r : route flow; E^{del} : delivered energy; E^{unsrv} : unserved energy; VOT = 1.0 \$/h. Source: `test_ev_power_traffic_report.cpp` (11 assertions, all passing).

Algorithm	h_1 (veh)	h_2 (veh)	E^{del} (kWh)	E^{unsrv} (kWh)	Travel (veh·h)	LP obj (\$)
DSP	5	0	40.0	60.0	2.50	—
Greedy	2	3	100.0	0.0	4.00	—
SO-LP	2	3	100.0	0.0	4.00	34.00

Remark 14 (SO/greedy vs. DUE-MSA). *DSP, Greedy, and SO-LP are single-iteration assignment methods, not DUE equilibrium solvers. They demonstrate energy delivery correctness and station-capacity enforcement, not Wardrop optimality. The CTM-MSA DUE solver is validated separately in Section 1.15.12 (4-node, 4 EVs, `test_ev_power_traffic_ctm_due.cpp`, 15 test cases, 159 assertions passing), which reports final EV Wardrop gap ϵ^E , ICV gap ϵ^F , session synthesis, energy delivery, schedule-delay route reversal, and price-of-anarchy.*

DSP station overload. The Deterministic Shortest Path assigns all 5 vehicles to the near route, ignoring station capacity. Station 101 can supply only 40 kW (2 plugs $\times$ 20 kW/plug), so 2 of 5 vehicles receive their full 20 kWh each (= 40 kWh delivered) and 3 vehicles receive nothing (= 60 kWh unserved). A queue forms: 5 vehicles are waiting in queue at step 2 (average wait 1.75 h), 3 at step 3 (1.05 h), 1 at step 4 (0.35 h), confirming the $M/D/c$ -type queue dynamics of Section 1.9.

Greedy and SO-LP agreement. Both the Capacity-Aware Greedy and the HiGHS-certified System-Optimal LP produce the identical split ($h_1 = 2$, $h_2 = 3$), saturating both stations at exactly their installed capacity (40 and 60 kW respectively) for one step each with zero queue. Table 5 shows the per-step station states for the Greedy case; the SO-LP results are numerically identical.

Table 5: Per-step station state — Capacity-Aware Greedy and SO-LP (identical). P_s^{ch} : station charging power (kW); B_s : vehicles actively charging; H_s : vehicles in post-service dwell awaiting egress. At step 2, $B = 0$ and $H > 0$ because all charging completed at step 1; no pre-service waiting ($Q_s = 0$ throughout).

Step k	P_{101}^{ch} (kW)	B_{101} (veh)	H_{101} (veh)	P_{102}^{ch} (kW)	B_{102} (veh)	H_{102} (veh)
0	0	0	0	0	0	0
1	40	2	0	60	3	0
2	0	0	2	0	0	3
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

System-Optimal LP objective. The LP minimises total system cost; with station price 0.30\$/kWh and VOT = 1.0\$/h:

$$J^{SO} = \underbrace{0.30 \times 100}_{\text{energy cost}=\$30.00} + \underbrace{1.0 \times (2 \times 0.5 + 3 \times 1.0)}_{\text{travel delay}=\$4.00} = \$34.00. \quad (91)$$

The LP is proven optimal by HiGHS (NativeDualSimplex backend), MIP gap = 0 (pure LP, no integer variables), primal infeasibility $< 10^{-14}$.

1.15.12 CTM-DUE MSA validation: 4-node benchmark

Setup. The same 4-node, 4-link symmetric network of Section 1.15.11 is used, but solved with the full CTM-MSA DUE algorithm (eq:ctm-due-algorithm). 4 EVs depart at step 0, each requesting 10 kWh at station 101 (route 1) or station 102 (route 2); $\Delta t = 1$ h, $T = 5$, $\epsilon^{tol} = 10^{-4}$, max 50 iterations. Source: `test_ev_power_traffic_ctm_due.cpp` (15 test cases, 159 assertions, all passing; B1–B3 outputs below).

B1: Session synthesis and dispatch.

Metric	Value
Converged	yes
MSA iterations	1
ϵ^E (final EV Wardrop gap, eq:ev-gap)	0.00×10^0
ϵ^F (final ICV Wardrop gap, eq:icv-gap)	0.00×10^0
Sessions synthesised	1
E^{req} (total requested energy)	40.00 kWh
E^{del} (total delivered energy)	40.00 kWh
E^{unsrv} (unserved energy)	0.00 kWh
Service ratio	100.0%
Route 1 flow h_1	4.000 veh
Route 2 flow h_2	0.000 veh
Peak station 101 load (step 1)	42.1 kW
Station 101 queue	0 veh (all steps)

On an uncongested symmetric network with no schedule delay all routes have equal generalised cost at the initial equal-split, so the relative gap $\epsilon^E = 0$ at iteration 1. The auxiliary AoN assignment concentrates flow on route 1 (tie-broken by index); the MSA step produces $h_1 = 4$, $h_2 = 0$. All 4 vehicles are served without queueing ($W_{s,k} = 0$ hr for all s, k), and the single synthesised session delivers all 40 kWh.

B2: SOC feasibility filter. Route 1 requires $2 \times 0.5 \text{ km} \times 2 \text{ kWh/km} = 2.0 \text{ kWh}$ driving energy; route 2 requires $2 \times 1.5 \times 2 = 6.0 \text{ kWh}$. With $e_0 = 2.0 \text{ kWh}$, route 2 is infeasible (eq:soc-feasibility-filter) and is removed before AoN assignment: $h_1 = 4.0000$ veh, $h_2 = 0.000000$ veh. Total flow conserved: $h_1 + h_2 = 4.0$.

B3: Price-of-anarchy benchmark. $\Delta t = 0.1$ h, $T = 6$; both DUE and SO benchmarks run.

Metric	Value
Converged	yes
MSA iterations	1
ϵ^E (final EV gap)	0.00×10^0
DUE TSTT	0.4000 veh·hr
SO TSTT	0.0000 veh·hr
PoA ratio	1.00
SO benchmark solved	yes

At free-flow speed on an uncongested network the DUE TSTT equals $4 \text{ veh} \times 0.10 \text{ hr/route} = 0.40 \text{ veh} \cdot \text{hr}$. The SO benchmark produces TSTT = 0 because the system-optimal LP minimises total link flow weighted by travel time; with zero congestion (all BPR terms at base value) the LP finds a trivial lower bound of 0. The PoA ratio = DUE TSTT/SO TSTT is reported as 1.0 when the denominator is below 10^{-9} (both values are near free-flow; no congestion to exploit).

Schedule-delay reversal (test case 11). To validate eq:psi-icv and eq:icv-schedule-delay, the asymmetric network (route 1: ff ≈ 0.10 hr; route 2: ff ≈ 0.30 hr) is run with an ICV demand (6 vehicles) whose desired arrival time is $t^* = 0.25$ hr and early penalty $\gamma_e = 20$ \$/hr, $\gamma_l = 0$.

Without schedule delay, route 1 wins (shorter travel time). With the penalty:

$$\Psi_1^F = 0.10 + \frac{\gamma_e \max(0, 0.25 - 0.10)}{VOT} = 0.10 + 20 \times 0.15 = 3.10 \$, \quad \Psi_2^F = 0.30 + 0 = 0.30 \$.$$

AoN assigns all 6 ICV vehicles to route 2 (verified: $h_2 > h_1$, total = 6.0 veh, $\epsilon^F \geq 0$ and finite).

Gap fields (test case 12). On a symmetric EV-only problem converged to $\epsilon^{tol} = 10^{-3}$: $\epsilon^F = 0$ (no ICV demands), $\epsilon^E \geq 0$, finite, and $\epsilon^E < \epsilon^{tol}$ upon convergence; `CTMDUEResult::relative_gap` = $\max(\epsilon^E, \epsilon^F)$ (all verified by 4 assertions).

1.15.13 Joint social welfare optimizer: FormD small case (2-bus, 4 EVs)

Remark 15 (FormD solver modes and certificates). *The default Formulation D solver (`joint_optimizer.cpp`) is the solver-certified LP/MILP reported in this numerical table. It uses either free-flow travel times, or one exogenous CTM pre-pass when `use_ctm_travel_times=true`; in both cases the LP/MILP has a valid primal feasibility residual, dual bound, and MIP gap for the assembled linear/integer model. A new `CertifiedDynamicMPECMILP` / `CertifiedDynamicUserBenefitMPECMILP` mode assembles a finite linear dynamic MPEC MILP with endogenous affine link costs, road-fault choice filtering, Wardrop big-M complementarity, charging/V2G, and optional DC-OPF in one solver call. Its global certificate is the MILP certificate for that finite linear model, not a proof for nonlinear CTM/LTM. The newer `CertifiedFullJointLtmPwlMILP` / `CertifiedFullJointLtmUserBenefitPwlMILP` mode keeps the same charging/V2G, DC-OPF, and Wardrop big-M rows but adds LTM cumulative count variables $N_{a,k}^{in}, N_{a,k}^{out}$, LTM sending/receiving and availability-capacity rows, and a piecewise-linear BPR travel-cost envelope. Its certificate is a global MILP certificate for the assembled LTM PWL-MILP approximation, with `pwl_max_abs_error_bound` reporting the travel-cost approximation boundary. It is explicitly not an exact global certificate for the original nonlinear CTM/LTM MPEC. A separate monolithic nonlinear mode, `JointOptimizerMode::FullJointSocialWelfareNLP` / `FullJointUserBenefitNLP`, is implemented in `joint_optimizer_full_nlp.cpp`. That mode assembles endogenous BPR congestion costs, Wardrop complementarity rows, aggregate charging/V2G and SOC equations, and optional DC-OPF equations in one reduced finite-route NLP/MPEC. Its certificate is local NLP feasibility/stationarity only; it is not a global nonlinear MIP-gap proof for the full CTM/LTM MPEC.*

Network and parameters. A 2-bus DC power system (bus 1: generator G1 at 10 \$/MWh, $\bar{p}_1^g = 2$ MW; bus 2: generator G2 at 50 \$/MWh, $\bar{p}_2^g = 0.5$ MW; branch 1-2: $x = 0.1$ pu, rate = 0.5 MW). Traffic sub-network identical to Section 1.15.11 (4 nodes, 2 routes). 4 EVs, WTP = \$50/veh, energy request = 10 kWh/veh, VOT = 1.0 \$/h, station price = 0.03 \$/kWh; $\Delta t = 1$ h, $T = 6$. Station 101 on bus 1 (40 kW, 2 plugs); station 102 on bus 2 (60 kW, 3 plugs).

Table 6 reports three LP/MILP solver modes from `test_ev_power_traffic_joint_opt_d.cpp`. The same test file also contains a certified dynamic MPEC MILP case, local-certificate checks for the full-joint nonlinear mode, and honesty tests that reject unsupported global nonlinear certificates.

Certified finite dynamic MPEC MILP numerical check. The D14 regression case activates `CertifiedDynamicUserBenefitMPECMILP` with `require_exact_mathematical_model=true`. Link 11 is unavailable at departure step 0, so the near route is removed from the feasible route-departure set. The remaining far route includes a V2G-capable stop at station 102, and bus 2 carries a small base load so reverse EV power can be absorbed by the DC-OPF balance. HiGHS returns an optimal MILP solution with zero reported MIP gap, primal residual below

Table 6: FormD joint optimizer: 2-bus, 4-EV results. SWmax = SocialWelfareMax; UBM = UserBenefitMax; Vars/Cons: LP variable/constraint counts; W : social welfare.

Mode	h_1 (veh)	h_2 (veh)	B_{EV} (\$)	C_{del} (\$)	C_{gen} (\$)	W (\$)	Vars / Cons
SWmax, no OPF	4.0	0.0	200.00	2.00	0.00	198.00	11 / 27
SWmax + DC-OPF	4.0	0.0	200.00	2.00	0.40	197.60	53 / 45
UBM, no OPF	4.0	0.0	200.00	2.00	0.00	—	11 / 27

10^{-6} , and integrality residual below 10^{-6} . The result fields are

```
certified_dynamic_mpec_model=true, travel_times_are_endogenous=true, wardrop_complementarity_residual=0.0,
dcopf_coupling_enforced=true, charging_v2g_enforced=true, global_optimum_certificate=1
```

The numerical route flow is $h_1 = 0$, $h_2 = 4$, with all four vehicles served. The Wardrop complementarity residual and road-capacity violation are both below test tolerance. This verifies the assembled finite linear MILP/MPEC equations; it does not re-label the nonlinear FullJoint NLP as a global optimizer.

Certified full-joint LTM PWL-MILP numerical check. The D15 regression case activates `CertifiedFullJointLtmUserBenefitPwlMILP` with `require_exact_mathematical_model=true`. The case uses the same road-failure and V2G setting as D14, but the solver model also contains LTM cumulative-count variables, link monotonicity, sending/receiving, entry/exit-capacity, and availability rows. The nonlinear BPR link-cost curve is represented by a piecewise-linear envelope with six segments; the maximum segment error is reported through `pwl_max_abs_error_bound`. HiGHS returns a globally certified MILP solution satisfying the configured MIP gap, primal residual, integrality residual, Wardrop complementarity residual, and `ltm_conservation_max_violation` tolerances. The asserted result fields include

```
certified_full_joint_ltm_pwl_milp_model=true, full_ltm_dynamics_enforced=true, pwl_max_abs_error_bound=0.0,
travel_times_are_endogenous=true, dcopf_coupling_enforced=true, charging_v2g_enforced=true
```

The route-flow outcome is again $h_1 = 0$, $h_2 = 4$, with positive V2G energy and populated DC-OPF dispatch. This is the strongest current certified path for the joint model, but its honesty boundary is essential: the certificate is for the LTM PWL-MILP approximation, not for the exact nonlinear CTM/LTM-DUE/MPEC.

Full-joint nonlinear mode. For a finite route/departure set $\mathcal{A}_d$, the implemented monolithic mode introduces choice flows h_i and demand minimum-cost variables μ_d . Endogenous route costs are computed from BPR link costs,

$$C_i(h) = \text{VOT}_d \sum_{a \in r_i} t_a^0 \left(1 + \alpha_a \left(\frac{x_{a,k_i}}{\bar{x}_{a,k_i}} \right)^{\beta_a} \right) + C_i^{chg} + C_i^{sch}, \quad x_{a,k} = \sum_{i: a \in r_i, k_i=k} h_i. \quad (92)$$

Wardrop equilibrium is assembled as complementarity equations and non-negativity inequalities:

$$C_i(h) - \mu_d \geq 0, \quad h_i(C_i(h) - \mu_d) = 0, \quad i \in \mathcal{A}_d. \quad (93)$$

The implementation supplies analytic sparse derivatives to the local NLP solver. For link-flow sensitivity,

$$\frac{\partial C_i}{\partial h_j} = \left(\text{VOT}_d + \frac{\partial C_i^{sch}}{\partial t_i^{arr}} \right) \sum_{a \in r_i \cap r_j: k_i=k_j} t_a^0 \alpha_a \beta_a \frac{1}{\bar{x}_{a,k_i}} \left(\frac{x_{a,k_i}}{\bar{x}_{a,k_i}} \right)^{\beta_a-1}. \quad (94)$$

Thus the Wardrop-product row has Jacobian entries

$$\frac{\partial}{\partial h_j} [h_i(C_i - \mu_d)] = \mathbf{1}_{i=j}(C_i - \mu_d) + h_i \frac{\partial C_i}{\partial h_j}, \quad \frac{\partial}{\partial \mu_d} [h_i(C_i - \mu_d)] = -h_i. \quad (95)$$

The same NLP contains demand conservation, road-capacity inequalities, station charge/discharge bounds, SOC recursions, terminal energy requirements, and, when `include_dcopf=true`, DC branch-flow and nodal power-balance equations with EV net load $p_{b,k}^{EV} = 10^{-3} \sum_{s: \text{bus}(s)=b} (P_{s,k}^{ch} - P_{s,k}^{dis})$. These linear blocks are inserted directly into the sparse Jacobian. When `full_joint_verify_sparse_derivatives=true`, the code compares the analytic gradient, equality Jacobian, and inequality Jacobian against finite-difference values at the assembled initial point and reports `sparse_derivatives_verified`, `derivative_check_max_abs_error`, and `derivative_check_failures`.

The returned `JointOptimizerResult` marks `monolithic_full_joint_model=true`, `endogenous_congestion_enforced=true`, and `wardrop_complementarity_enforced=true`. It deliberately keeps `global_optimum_certificate=false` and `proven_optimal=false` unless a future global MPEC/MINLP solver supplies a valid dual bound. If `require_global_nonlinear_certificate=true`, the implementation returns an unsupported status instead of silently treating the sparse NativeNLP local solve as a global proof.

Welfare accounting. All four vehicles prefer the near route: WTP = \$50/veh exceeds both the travel delay ($0.5 \text{ h} \times \$1/\text{h} = \$0.50/\text{veh}$) and the energy cost ($10 \text{ kWh} \times 0.03 \text{ $/kWh} = \$0.30/\text{veh}$). Station 101 delivers $4 \times 10 = 40 \text{ kWh}$ in one step at $P_{101}^{ch} = 4 \times 10 \text{ kW} = 40 \text{ kW}$.

$$B_{EV} = 4 \times \$50 = \$200. \quad (96)$$

$$C_{del} = \text{VOT} \times (4 \times 0.5 \text{ h}) = \$2.00. \quad (97)$$

$$C_{gen} = 0 \quad (\text{no OPF coupling in SWmax no-OPF mode}). \quad (98)$$

$$W = B_{EV} - C_{del} - C_{gen} = 200 - 2 - 0 = \$198.00. \quad (99)$$

With DC-OPF coupling, generator G1 dispatches 0.04 MW at step 1 to balance the EV load:

$$C_{gen} = 10 \text{ $/MWh} \times 0.04 \text{ MW} \times 1 \text{ h} = \$0.40, \quad W = 200 - 2 - 0.40 = \$197.60. \quad (100)$$

The DC-OPF formulation adds 53 LP variables (generators, voltage angles, branch flows for all T steps) and 45 constraints versus the 11 variables and 27 constraints of the traffic-only LP.

For `UserBenefitMax` the LP minimises $C_{del} - B_{EV} + C_{elec}$ where $C_{elec} = 4 \times 10 \text{ kWh} \times 0.03 \text{ $/kWh} = \$1.20$, giving objective = $2.00 - 200.00 + 1.20 = -\196.80 .

Solver certificate. All three modes: proven optimal, solver status `Optimal`, `NativeDualSimplex` backend, primal infeasibility $\leq 7 \times 10^{-11}$, integrality violation = 0 (pure LP).

1.15.14 V2G pure-LP validation (Proposition 1)

Setup. One EV ($n_{vehicles} = 1$) on route 1 only; $e_0 = 50 \text{ kWh}$, $e_{min} = 20 \text{ kWh}$, $e_{max} = 60 \text{ kWh}$; dwell = 2 steps; station price $\pi_s = 1.0 \text{ $/kWh}$ (high, to incentivise maximal discharge); $\bar{p}^{dis} = 10 \text{ kW}$, $\bar{p}^{ch} = 20 \text{ kW}$; WTP = \$50; VOT = 1.0 \$/h; mode: `UserBenefitMax` with V2G enabled. No binary variables are present (pure LP following Proposition 1).

Remark 16 (No-arbitrage condition in this test). *This test does not rely on Proposition 1 to enforce sequential behaviour. The EV begins with sufficient SOC and the objective rewards only delivered V2G energy; charging is never economically attractive, so the LP solution is discharge-only. The no-arbitrage condition (26) is not the reason no simultaneous charging*

occurs in this particular instance—it is a consequence of the LP cost structure under these specific parameters. The no-arbitrage condition $\pi^{dis}\eta^{ch}\eta^{dis} \leq \pi^{ch} + c^{deg}$ is a price-parameter condition: when it holds, a LP solution with $p_{v,k}^{ch}p_{v,k}^{dis} = 0$ is provably at least as good as any simultaneous schedule (Proposition 1). If instead $\pi^{dis}\eta^{ch}\eta^{dis} > \pi^{ch} + c^{deg}$ (arbitrage is profitable), binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ would be required.

Optimal V2G plan. Table 7 shows the LP-optimal per-step charging schedule.

Table 7: V2G pure-LP: per-step session plan for 1 EV (`test_ev_power_traffic_joint_opt_d.cpp`, test D9, all assertions passing).

Dwell step k	$p_{v,k}^{ch}$ (kW)	$p_{v,k}^{dis}$ (kW)	$e_{v,k\text{-start}}$ (kWh)	$e_{v,k\text{-end}}$ (kWh)	$p^{ch} \cdot p^{dis}$	Prop. 1
1	0.00	10.00	50	40	0	✓
2	0.00	10.00	40	30	0	✓

The product $p_{v,k}^{ch} \cdot p_{v,k}^{dis} = 0$ at both steps: the EV discharges at maximum 10kW for both dwell steps without any simultaneous charging. Under the no-arbitrage condition (26) (satisfied here: $\pi^{dis}\eta^{ch}\eta^{dis} = 1.0 \times 1 \times 1 \leq \pi^{ch} + c^{deg} = \infty$), the LP optimum is provably non-simultaneous by Proposition 1, so no binary mode constraints are required *for this test instance*. If the no-arbitrage condition were violated, binary constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ would be needed.

Objective decomposition. Total V2G energy = $2 \times 10 \text{ kW} \times 1 \text{ h} = 20 \text{ kWh}$; V2G revenue = $20 \text{ kWh} \times \$1.0/\text{kWh} = \20 ; travel delay $C_{del} = 1.0 \times 0.5 \text{ h} = \0.50 :

$$J^{UBM} = C_{del} - B_{EV} - E_{tot}^{dis} \cdot \pi_s = 0.50 - 50.00 - 20.00 \cdot 1.0 = -\$69.50. \quad (101)$$

Verified by test assertion: `res.objective` = -69.50 (tolerance 10^{-6}); `proven_optimal`=true; primal infeasibility $< 10^{-7}$.

Simulation cross-check. Test Scenario 4 in `test_ev_power_traffic_report.cpp` independently simulates one V2G session ($e_0 = 50 \text{ kWh}$, $p^{dis} = 20 \text{ kW}$, 1 dwell step): final SOC = 30 kWh , $E_{tot}^{dis} = 20 \text{ kWh}$, EV power balance margin increases by 0.020 MW at the V2G step. Both the simulation test and the joint-optimizer LP agree on $E^{dis} = 20 \text{ kWh}$.

1.15.15 Large-scale benchmark: 10×10 grid, 10,000 EVs, IEEE-118

System description.

- **Power grid:** IEEE 118-bus; buses 1–100 each host a fast-charging station with 300 fast plugs at 50 kW each (15 MW/station, \$0.30/kWh).
- **Traffic grid:** 10×10 Cartesian grid, 100 nodes, 180 directed links; free-flow 0.25 h/link, capacity 2000 veh/h.
- **Corridors:** 5 north–south corridors, each with two routes. Route A: 9 vertical links down the left column, 1 horizontal link at the bottom (charging stop at mid-left node, row 4). Route B: 1 horizontal link at the top, 9 vertical links down the right column (charging stop at mid-right node, row 4). $\Rightarrow$ 10 routes total.
- **Demands:** 5 corridors $\times$ 4 departure steps = 20 fleet demands, 500 vehicles/demand = 10 000 vehicles total. Each vehicle requests 30 kWh at up to 50 kW.
- $\Delta t = 1 \text{ h}$, $T = 12$; enforce road capacity and station capacity.

Algorithm comparison. Table 8 reports results from `test_ev_power_traffic_largescale.cpp` (14 assertions, all passing).

Remark 17 (Scalability vs. DUE validation). *The algorithms compared here (CapacityAwareGreedy and SystemOptimalLP) are assignment heuristics and a system-optimal LP, respectively; they are not DUE-MSA runs. This benchmark demonstrates scalability (10,000 EVs, 100 stations, IEEE-118 power grid) and energy-delivery correctness, not equilibrium convergence. A DUE-MSA run on this network would additionally report: final ICV gap ϵ^F , final EV gap ϵ^E , number of MSA iterations, departure-time split, and used-path cost dispersion. These metrics are omitted here because the test uses a single-iteration greedy/SO-LP approach for performance benchmarking.*

Table 8: Large-scale benchmark: 10×10 grid, 10,000 EVs, IEEE-118. All 10,000 vehicles served, $E^{del} = 300,000$ kWh, $E^{unsrv} = 0$.

Algorithm	Solve time (ms)	Travel time (veh·h)	Peak station (kW)	LP obj	Proved optimal
CapacityAwareGreedy	0.5	25,001	9,000	—	no
SystemOptimalLP	3.1	25,015	15,000	115,000	yes

Route assignment. Under CapacityAwareGreedy each of the 20 demands splits 300/200 (Route A/Route B), distributing load between mid-left stations (41, 43, 45, 47, 49) and mid-right stations (42, 44, 46, 48, 50). Under SystemOptimalLP all 500 vehicles per demand concentrate on Route A (LP objective = 115,000), loading left-column stations at their 15 MW installed capacity but achieving certified global optimality.

Station utilisation (CapacityAwareGreedy). Table 9 reports peak load and total delivered energy for the 10 route stations across all 4 departure steps.

Table 9: Per-station utilisation under CapacityAwareGreedy. Route-A stations (odd IDs) serve 300 vehicles/demand; Route-B stations (even IDs) serve 200 vehicles/demand; each station active for 4 departure-step groups.

Station (bus)	Route	Peak load (kW)	Delivered energy (kWh)
41	A, corridor 0	9,000	36,000
42	B, corridor 0	6,000	24,000
43	A, corridor 1	9,000	36,000
44	B, corridor 1	6,000	24,000
45	A, corridor 2	9,000	36,000
46	B, corridor 2	6,000	24,000
47	A, corridor 3	9,000	36,000
48	B, corridor 3	6,000	24,000
49	A, corridor 4	9,000	36,000
50	B, corridor 4	6,000	24,000
Total		—	300,000

Energy balance verified: $10,000 \times 30 \text{ kWh} = 300,000 \text{ kWh}$, matching `total_delivered_energy_kwh` to within 1 kWh. Route-A peak: each vehicle requests 30 kWh in a 1-hour step,

requiring an average charging power of $30 \text{ kWh}/1 \text{ h} = 30 \text{ kW}$ (within the 50 kW plug hardware limit); $300 \text{ veh} \times 30 \text{ kW/veh} = 9,000 \text{ kW}$. Route-B analogously: $200 \times 30 = 6,000 \text{ kW}$. Under SystemOptimalLP all 500 vehicles in each demand window are concentrated on Route A ($500 \times 30 = 15,000 \text{ kW}$, exactly saturating the 15 MW station capacity).

Scalability. Both solvers complete well within their declared time limits (CapacityAware-Greedy: 0.5 ms vs. 5 s limit; SystemOptimalLP: 3.1 ms vs. 30 s limit), demonstrating that the LP and greedy heuristic both scale efficiently to $10,000$ -vehicle fleets on a 100 -node grid.

1.16 Extension Interfaces to Power and Resilience Models

Output quantity	Consuming module
Station charging power $p_{s,k}^{ch}$	Smart-charging / DC-OPF (Section ??)
Station dwell population $H_{s,k}$ and per-EV SOC	V2G aggregator $\rightarrow$ OPF
EV departure flow $o_{s,k}$	LTM downstream receiving update
Aggregate load $P_{b,k}^{EV}$	Power-network load profile
V2G potential $P_{s,k}^{v2g,pot}$	Reserve and resilience models

1.17 Optimization Framework

The simulation sub-modules use MSA for DUE and rule-based heuristics for charging admission. This section gives proper mathematical programmes that replace or wrap those heuristic layers, providing optimality guarantees and enabling joint power-traffic problems.

Remark 18 (Implementation status of optimisation sub-modules). *Table 10 lists the source files. The default joint social welfare optimiser is a solver-certified LP/MILP over route, charging/V2G, per-step road-capacity, and optional DC-OPF rows. Its travel times are exogenous: either free-flow values or a one-shot equal-split CTM pre-pass. The separate FullJoint NLP mode is the only current path with endogenous congestion rows, and it is locally certified only. The remaining five formulations compile, expose direct public entry points, and are covered by `test_ev_power_traffic_formulations_e_f_g_h`. In the second implementation round, the optimisation modules were upgraded from static prototypes to solver-assembled dynamic programmes with explicit certificates:*

- *FormD full-joint NLP: a new monolithic reduced NLP/MPEC mode assembles endogenous BPR congestion, Wardrop complementarity, charging/V2G, and optional DC-OPF in one programme. It reports local feasibility/stationarity, uses analytic sparse derivatives, and explicitly refuses a requested global nonlinear certificate because the current in-tree solvers do not prove a global MPEC/MINLP gap for this model.*
- *SO-CTM LP: route-specific cell states, inter-cell flows, explicit node movement variables $y_{a,b,k}$, and aggregate merge/origin receiving constraints are assembled in the LP.*
- *SO-LTM LP: per-route cumulative counts are connected by explicit node movement variables; downstream receiving constraints aggregate all incoming movements and origin injection at the same link-step.*
- *DUE / Frank-Wolfe: full endogenous EV departure-window assignment is available through `DUEMode::FullEndogenous`; the final solution reports disaggregated departure-route flows, VI residual, normalised VI gap, complementarity residual, and demand-conservation residual computed at final CTM costs.*

- *Infrastructure MILP*: BPR Beckmann piecewise-linear cost is available; an optional time-expanded LTM dynamic envelope enforces route-link cumulative counts, node movement, capacity, and availability constraints inside the MILP.
- *LTM-MPC*: per-station access-link N_{in}/N_{out} tracking with strict SOC-V2G linkage; access-link inflows are bounded by an exogenous OD/route forecast so the controller cannot create artificial arrivals.

The remaining distinction is not merely cosmetic: DUE fixed points are checked by residual metrics, LP/MILP modules are certified only for their assembled linear/integer subproblems, and the nonlinear FullJoint mode has no global MPEC/MINLP proof. For paper-audit runs the C++ options expose `require_exact_mathematical_model`. When this flag is set, every reduced, residual-based, forecast-driven, aggregate, or locally certified path returns an `unsupported` status rather than a numerical result. Therefore a reported `mathematical_model_verified=true` cannot be produced by the present implementation unless a future solver path supplies an equation-by-equation full-model certificate.

1.17.1 CTM-based system-optimal traffic assignment (SO-CTM LP)

Under minimisation of total vehicle-time the non-smooth $\min(\cdot, \cdot)$ in (44) is linearised by two one-sided upper bounds (Ziliaskopoulos 2000).

Objective.

$$\min \sum_{a,m,k} n_{a,m,k} \Delta t. \quad (102)$$

Constraints.

$$n_{a,m,k+1} = n_{a,m,k} + y_{a,m-1,k} - y_{a,m,k} \quad \forall a, m, k \quad (103)$$

$$y_{a,m,k} \leq \frac{v_a^{ff}}{\delta_a} n_{a,m,k} \Delta t \quad \forall a, m, k \quad (104)$$

$$y_{a,m,k} \leq \bar{q}_a \Delta t \quad \forall a, m, k \quad (105)$$

$$y_{a,m,k} \leq \frac{w_a}{\delta_a} (N_{a,m+1}^{jam} - n_{a,m+1,k}) \Delta t \quad \forall a, m < M_a - 1, k \quad (106)$$

$$\sum_{r \in \mathcal{R}(od)} x_{r,k} = d_k^{od} \quad \forall od, k \quad (107)$$

$$n_{a,m,0} = 0 \quad \forall a, m \quad (108)$$

The initial empty condition $n_{a,m,0} = 0$ is standard. The terminal clearance constraint $n_{a,m,T} = 0$ is valid only when the planning horizon T is long enough that all demand has been served; for shorter horizons, replace it with a terminal penalty $\lambda_T \sum_{a,m} n_{a,m,T}$ in the objective (102).

The receiving constraint (106) applies to internal cell-to-cell flows ($m < M_a - 1$). For the exit flow from the last cell $m = M_a - 1$, the downstream receiving capacity is imposed by the node model of the successor link or sink: if link a discharges to a sink, the receiving capacity is set to the sink capacity (or ∞); for an ordinary downstream link the first-cell receiving constraint of that link provides the bound via the CTM node equations. LP size: $O((\sum_a M_a + |\mathcal{R}|)T)$ variables, $O(\sum_a M_a T)$ constraints.

1.17.2 LTM-based system-optimal assignment (SO-LTM LP)

Objective.

$$\min \sum_{a,k} (N_{a,k}^{in} - N_{a,k}^{out}) \Delta t = \min \sum_{a,k} n_{a,k} \Delta t, \quad (109)$$

the total vehicle-hours in the network (minimising total delay). The terminal residual $\sum_a (N_{a,T}^{in} - N_{a,T}^{out})$ may be used as an alternative when the planning horizon T is long enough that the network empties; it is easier to evaluate but ignores travel time throughout the simulation period.

Constraints (all linear).

$$N_{a,k}^{out} - N_{a,k-1}^{out} \leq N_{a,k-\tau_a^{ff}}^{in} - N_{a,k-1}^{out} \quad (\text{sending}) \quad (110)$$

$$N_{a,k}^{out} - N_{a,k-1}^{out} \leq \bar{q}_a \Delta t \quad (\text{cap}) \quad (111)$$

$$N_{a,k}^{in} - N_{a,k-1}^{in} \leq N_{a,k-\tau_a^{bw}}^{out} + N_a^{jam} - N_{a,k-1}^{in} \quad (\text{receiving}) \quad (112)$$

$$N_{a,k}^{in} - N_{a,k-1}^{in} \leq \bar{q}_a \Delta t \quad (\text{cap}) \quad (113)$$

$$\sum_b y_{a,b,k} = N_{a,k}^{out} - N_{a,k-1}^{out} \quad (\text{exit split}) \quad (114)$$

$$\sum_a y_{a,b,k} + d_{b,k}^{orig} = N_{b,k}^{in} - N_{b,k-1}^{in} \quad (\text{entry merge}) \quad (115)$$

$$N_{a,k}^{in} \geq N_{a,k-1}^{in}, \quad N_{a,k}^{out} \geq N_{a,k-1}^{out} \quad (\text{monotone}) \quad (116)$$

LP size: $O(|\mathcal{A}|T)$, independent of cell count.

1.17.3 DUE as a variational inequality

Let the feasible departure-rate set be

$$\mathcal{H} = \left\{ (h^F, h^E) \geq 0 : \sum_{k \in \mathcal{K}_w^{dep}} \sum_r h_{w,r,k}^F = D_w^F, \sum_{k \in \mathcal{K}_w^{dep}} \sum_\omega h_{w,\omega,k}^E = D_w^E, \forall w \right\}. \quad (117)$$

The full-DUE feasible set $\mathcal{H}$ from (117) contrasts with the fixed-departure feasible set:

$$\mathcal{H}^{fix} = \left\{ (h^F, h^E) \geq 0 : \sum_r h_{w,r,k}^F = d_{w,k}^F, \sum_\omega h_{w,\omega,k}^E = d_{w,k}^E, \forall w, k \right\}. \quad (118)$$

The mixed-fleet DUE condition (81)–(84) is equivalent to the variational inequality: find $(h^{F,*}, h^{E,*}) \in \mathcal{H}$ such that

$$\sum_{w,r,k} \Psi_{w,r,k}^F(h^*)(h_{w,r,k}^F - h_{w,r,k}^{F,*}) + \sum_{w,\omega,k} \Psi_{w,\omega,k}^E(h^*)(h_{w,\omega,k}^E - h_{w,\omega,k}^{E,*}) \geq 0 \quad \forall (h^F, h^E) \in \mathcal{H}. \quad (119)$$

For static or separable link-cost approximations the VI can be reduced to the Beckmann potential

$$\min_{(h^F, h^E) \in \mathcal{H}} \Phi(h) = \sum_a \int_0^{f_a(h)} t_a(u) du \quad (120)$$

which is convex when link cost functions are monotone and separable, and can be solved by Frank-Wolfe or interior-point methods. In the full dynamic CTM/LTM setting with spill-back and station queues, path costs Ψ^F, Ψ^E depend on departure time and link costs are non-separable; the DUE is therefore treated as the fixed-point problem (85) or the VI (119) rather than a Beckmann program. The Beckmann formulation and its uniqueness certificate are available only under the separable-monotone static approximation.

1.17.4 Joint power-traffic social welfare optimisation

$$\max_{x, p^g, p^{ch}} \sum_d W_d^P s_d - \sum_g C_g(p^g) - \text{VOT} \sum_r x_r T^r(x), \quad (121)$$

subject to DC power balance, LTM traffic dynamics, and charging-load coupling. The general coupling is expressed through station net load:

$$p_{b,k}^{EV} = \frac{1}{1000} \sum_{s: \text{bus}(s)=b} (P_{s,k}^{ch} - P_{s,k}^{dis}) \quad [\text{MW}], \quad (122)$$

where

$$P_{s,k}^{ch} = \sum_{\omega \ni s} x_{\omega,k}^E p_{\omega,k}^{ch}, \quad P_{s,k}^{dis} = \sum_{\omega \ni s} x_{\omega,k}^E p_{\omega,k}^{dis}. \quad (123)$$

Remark 19 (P^{EV} sign convention). *In this traffic module, positive $p_{b,k}^{EV}$ denotes net EV load (aggregate charging exceeds discharging). In power-flow equations using injection-positive convention, EV load enters as $-p_{b,k}^{EV}$, so that generation equals load plus net EV demand.*

In simplified benchmark cases without V2G and with all admitted vehicles charging at a fixed plug power, this reduces to

$$p_{b,k}^{EV} = \frac{1}{1000} \sum_{s: \text{bus}(s)=b} \sum_{r \ni s} x_{r,k} \bar{p}_s^{ch}. \quad (124)$$

The current benchmark tests use this simplified form; the station net-load form (122) is the mathematical interface for full V2G coupling.

Remark 20 (Implemented solver: LP/MILP, certified finite MILP/MPEC, and reduced full-joint NLP modes). *The C++ implementation in `joint_optimizer.cpp` (`solve_joint_optimizer`) supports the solver-certified LP/MILP modes described above. In the default static mode, free-flow route travel times T_r^{FF} are used; when `JointOptimizerOptions::use_ctm_travel_times=true`, one CTM forward pass supplies exogenous $T_{r,k_{dep}}^{CTM}$ before the LP/MILP is assembled. The `CertifiedDynamicMPECMILP` and `CertifiedDynamicUserBenefitMPECMILP` modes add a finite linear dynamic MPEC MILP with affine endogenous congestion, availability/fault filtering, Wardrop big-M rows, charging/V2G, and optional DC-OPF; its global certificate is the MILP certificate for that finite model. In addition, `joint_optimizer_full_nlp.cpp` implements a reduced monolithic nonlinear NLP/MPEC with endogenous BPR congestion, charging/V2G, DC-OPF, and Wardrop complementarity in one programme. This nonlinear mode is an exact assembly of the finite-route reduced equations, with analytic sparse derivatives and optional finite-difference derivative verification, but the current in-tree solve path reports only a local NLP feasibility/stationarity certificate. It is not a global certificate for the full CTM/LTM nonlinear MPEC.*

1.17.5 MILP for charging infrastructure design

Let $z_s \in \{0, 1\}$ and $\bar{B}_s \in \mathbb{Z}_+$:

$$\min_{z, \bar{B}, x, n} \sum_s (c_s^{cap} \bar{B}_s z_s + c_s^{op} \sum_k p_{s,k}^{ch} \Delta t) + \text{VOT} \sum_r x_r T^r, \quad (125)$$

with $B_{s,k} \leq \bar{B}_s z_s$ and $\bar{B}_s \leq \bar{B}_s^{max} z_s$ (big-M linking constraints). The implemented solver uses static OD conservation, station power/plug constraints, optional BPR link-flow variables, and native branch-and-cut through `engine::solve_milp_bc`. When `InfraDesignMILPOptions::include_ltm_dynamic_constraints` is enabled, the MILP also includes time-expanded LTM cumulative-count variables, node movement variables, link capacity/availability constraints, and downstream merge receiving constraints. This gives a solver-certified dynamic design envelope; a full CTM cell-state infrastructure MILP is intentionally left out because its variable count grows with $\sum_a M_a T$.

1.17.6 LTM-based model predictive control (MPC)

At each step k , solve a finite-horizon LP over horizon H with stage cost

$$\ell_{k+j} = \omega_Q \sum_s Q_{s,k+j} + \omega_T \sum_a (N_{a,k+j}^{\text{in}} - N_{a,k+j}^{\text{out}}) - \omega_{V2G} \sum_s P_{s,k+j}^{v2g,pot}, \quad (126)$$

controlled via admission rates $b_{s,k}, \dots, b_{s,k+H-1}$. Only the first-step action $b_{s,k}$ is applied; the LP is resolved at step $k+1$ with the observed state.

Recommended integration sequence.

1. **Offline design:** SO-CTM LP or infrastructure MILP to select station locations, capacities, and base route flows.
2. **Day-ahead:** joint social welfare optimisation (`solve_joint_social_welfare`) to set station prices $\pi_{s,k}$ aligning EV charging with grid marginal costs.
3. **Real-time:** LTM-MPC adjusting admission rates in response to observed queue lengths and grid emergency signals.

Remark 21 (Residual gaps between code and mathematical model). *All five formulations have direct regression tests (`test_ev_power_traffic_formulations_e_f_g_h.cpp`) that verify LP/MILP feasibility, primal-certificate residuals, node-flow conservation, and demand conservation at the solver-returned solution. The following gaps between the exact mathematical model and the current implementation remain:*

- **SO-CTM LP and SO-LTM LP — aggregate vs. per-route state representation.** `CTMOptions::use_aggregate_link_variables` (and the corresponding `LTMOptions` flag) switches the LP to reduced aggregate link/cell state variables $n_{a,m,k}$ and $N_{a,k}^{\text{in/out}}$ shared across routes. Per-route entry/exit or node-movement variables are still retained for OD accounting, so this is an aggregate relaxation/approximation of the full route-commodity model, not a proof that route-level FIFO is exactly enforced. When the flag is `false` (default), the per-route multi-commodity decomposition is used instead. Result fields `per_route_decomposition_used`, `closed_link_handling_strict`, and the LTM aggregate strictness flags expose which scope was actually solved.
- **SO-CTM LP — CFL stability auto-enforced (resolved).** The Courant–Friedrichs–Lewy ceiling $M_a^{\text{CFL}} = \lfloor \ell_a / (v_a^{\text{ff}} \Delta t) \rfloor$ is now computed per link and the user-supplied `CTMOptions::n_cells_per_link` is clamped to $\min(M_{\text{user}}, M_a^{\text{CFL}})$ at setup time. When the user supplies an explicit $\Delta t_{\text{CTM}} > 0$ the clamp applies to that value; otherwise Δt_{CTM} is auto-computed from `choose_ctm_dt` with a $0.9 v^{\text{ff}}$ safety margin that already satisfies CFL.
- **DUE-VI — mixed-fleet certificate scope.** `CTMDUEVIOptions::include_icv_in_due` enables ICV participation in the Frank–Wolfe loop. When set, both EV and ICV route flows are updated via auxiliary-assignment steps (`auxiliary_assignment_icv`) and convergence is checked using the worst EV/ICV Wardrop gap available at the final CTM costs. However, `compute_due_vi_certificate` still computes the VI residual over EV demand variables only. Therefore, when ICVs participate, `CTMDUEResult::vi_certificate_covers_ev_only=true`; the ICV side is reported through `icv_relative_gap`, not a full mixed-fleet VI dual certificate.

Table 10: C++ extension map for optimisation-based traffic models. **Implemented**: source file compiles, links, and has a direct regression test. **Implemented[†]**: source exists but approximates the exact mathematical formulation described in this chapter.

Formulation	Source file	Notes	Status
Joint SW optimizer	<code>joint_optimizer.cpp</code> / <code>assignment_lp.cpp</code>	Free-flow (default) or CTM-dynamic travel times (<code>use_ctm_travel_times</code>); per-step road-capacity bounds; HiGHS LP; DC-OPF optional	Implemented
SO-CTM LP	<code>ctm_so_lp.cpp</code>	<code>engine::LPModel</code> , HiGHS; multi-commodity per-route cell states $n_{a,m,k}^{(r)}$ with aggregate capacity constraints (equations (103)–(106) use the aggregate single-commodity $n_{a,m,k}$); flexible origin injection; link-availability support; explicit node movement variables and aggregate downstream receiving constraints; $M = \max(1, \text{n_cells_per_link})$ is user-supplied and is not auto-enforced for CFL stability	Implemented[†]
SO-LTM LP	<code>ltm_so_lp.cpp</code>	<code>engine::LPModel</code> , HiGHS; multi-commodity per-route cumulative counts $N^{\text{in},(r)} / N^{\text{out},(r)}$ with aggregate node capacity/receiving constraints (equations (110)–(112) use link-aggregate variables); correct origin injection; link-availability support; explicit $y_{a,b,k}$ movement variables (per-route <code>Ynode</code>) and merge receiving constraints; <code>so_tsstt_hr</code> = LP objective	Implemented[†]
DUE / Frank-Wolfe	<code>ctm_due_vi.cpp</code>	Frank-Wolfe with CTM forward pass; EV-demand only (eq. (119) includes mixed ICV+EV terms $\Psi^F + \Psi^E$; ICV traffic handled separately via <code>simulate_icv_traffic</code>); golden-section line search; departure-window support; Wardrop/VI residual stopping metrics; re-	Implemented[†]

- **Infrastructure MILP — BPR objective approximation.** The objective (125) specifies $\text{VOT} \sum_r x_r T^r(x)$ with endogenous travel times. The code replaces this with a piecewise-linear BPR Beckmann approximation; the excess-flow slope $\alpha\beta = 0.6$ matches the first-order derivative at $v_a = \bar{q}_a$. When `include_ltm_dynamic_constraints` is enabled, the $N_{a,k}^{\text{in}}/N_{a,k}^{\text{out}}$ variables receive objective coefficients $+\text{VOT} \cdot \Delta t$ and $-\text{VOT} \cdot \Delta t$ respectively, so their difference $N_{a,k}^{\text{in}} - N_{a,k}^{\text{out}}$ proxies vehicle-hours in transit; the previously double-counted static route-cost h -variable and BPR link-flow terms are zeroed out when the LTM block is active. Y_{node} movement-flow variables carry zero cost (they are kinematic feasibility variables).
- **Certified dynamic MPEC MILP — global certificate scope.** The `CertifiedDynamic*MPECMILP` modes are globally certified only for their finite linear model: affine link-delay costs, admitted route-departure choices, big-M Wardrop rows, charging/V2G, and optional DC-OPF. They deliberately do not claim exact nonlinear CTM cell propagation, LTM FIFO reconstruction, nonlinear BPR powers, or a global certificate for `FullJoint*NLP`. The Wardrop cost uses route-level planned charging/V2G energy terms; optimized session powers are coupled through linear SOC and station rows, but their realized power-per-served-vehicle ratios are not placed back into the Wardrop cost, because that would create bilinear equations outside the certified MILP class. Therefore, `mathematical_model_verified=true` in these modes means “verified for the assembled finite MILP/MPEC”, not “the whole paper-level nonlinear CTM/LTM MPEC has been globally solved”.
- **LTM-MPC — tracked-link VHT proxy.** `LTMPCOptions::track_all_route_links` extends the MPC LP with $N_{a,k}^{\text{in}}/N_{a,k}^{\text{out}}$ variables for every link on each station’s approach route. The ω_T stage cost then sums over all tracked links, which is closer to stage cost (126) than the single-access-link proxy. The admission constraint remains anchored to the final access link only. When the flag is `false` (default), only the single access-link proxy is used. The OD-forecast arrival bound continues to use a static free-flow proportional allocation (`forecast_station_arrivals_from_demands`), not a dynamic traffic prediction.